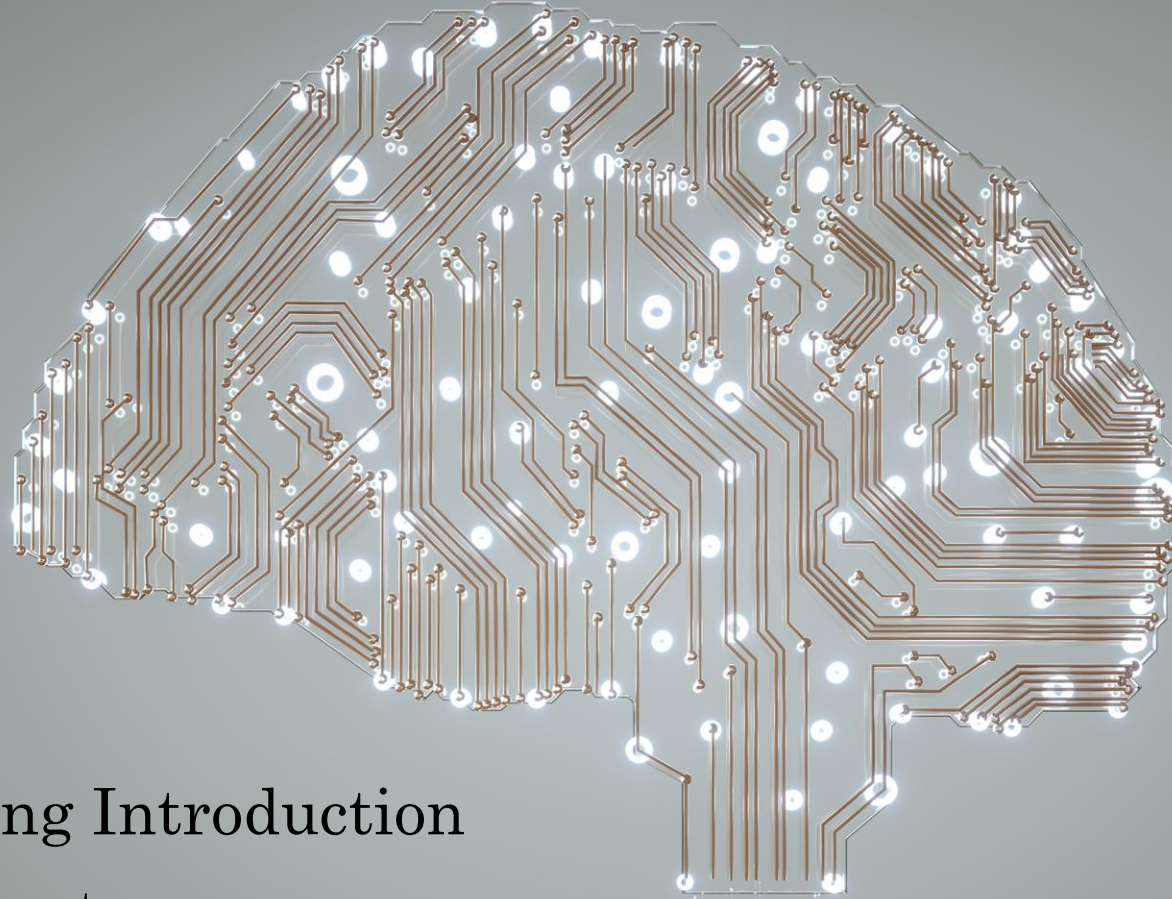
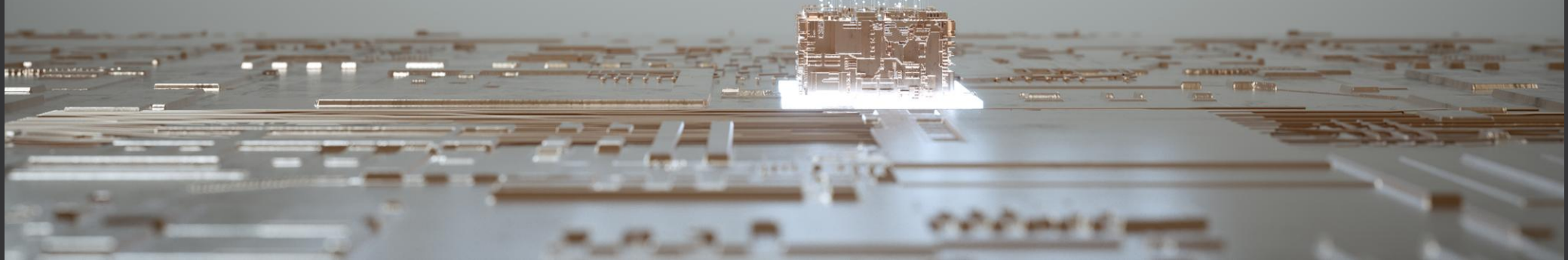


ML & NLP: PGR210



Deep Learning Introduction

Dr. Arvind Keprate



Topics for Today

What is Deep Learning

ML vs. DL

Why is DL so popular now?

Introduction to Neural Network

Forward and Backward Propagation

Gradient Descent Algorithm

Activation function

Loss Function

Questions worth wondering about?

- How does an autonomus car drive?



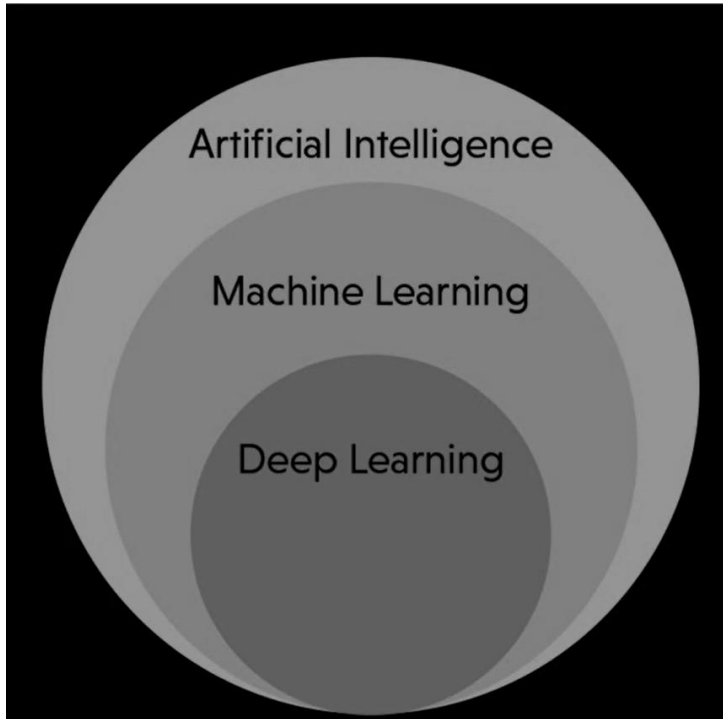
- How does Siri respond to your voice commands?

- It's not MAGIC, it's DEEP LEARNING

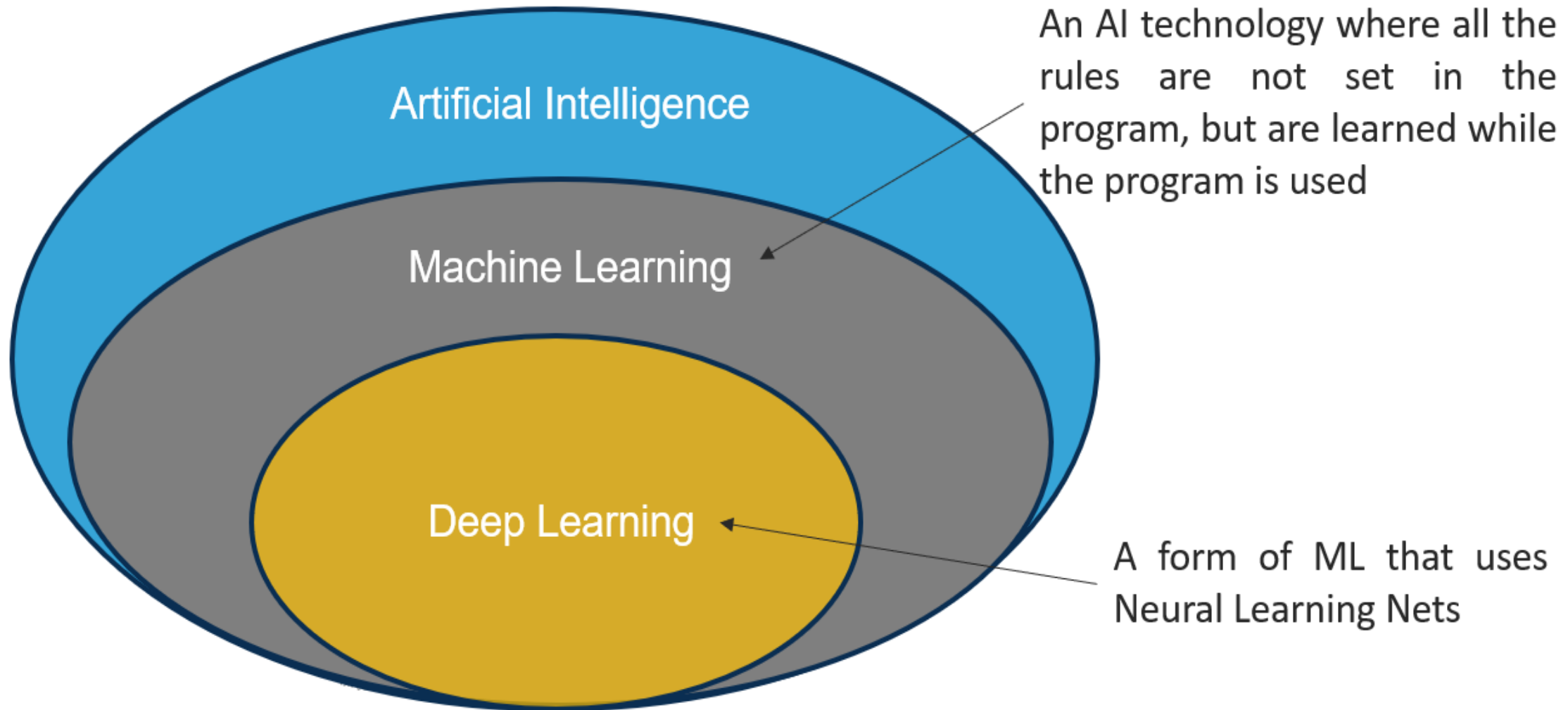
What is Deep Learning?

Did you know:

- Deep Learning is a sub class of Machine Learning.
- Uses Artificial Neural Networks (ANN) to learn from data.
- Artificial Neuron is inspired from the structure of human brain neuron.



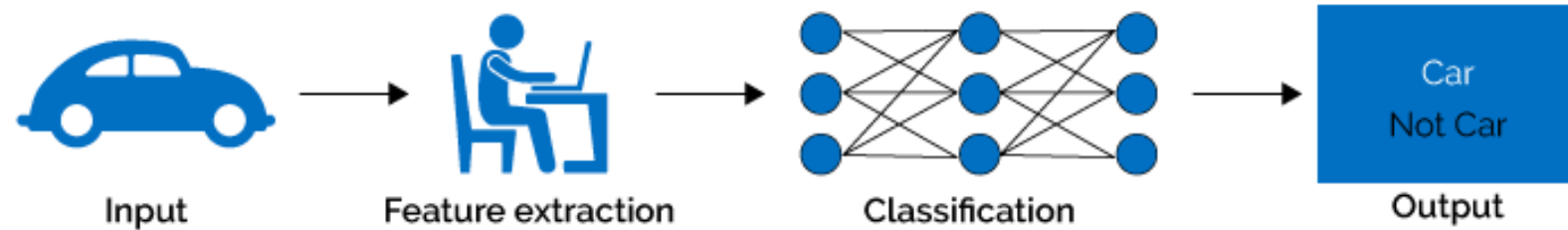
What is Deep Learning?



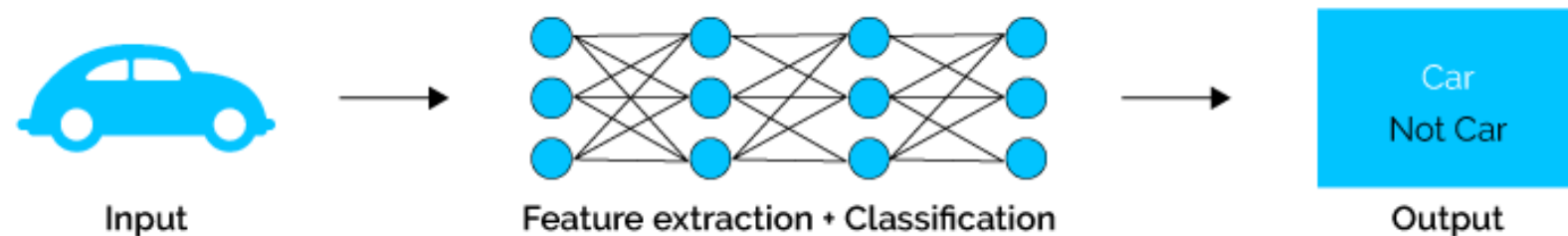
Machine Learning vs. Deep Learning

Automatic
Feature
Extraction

Machine Learning

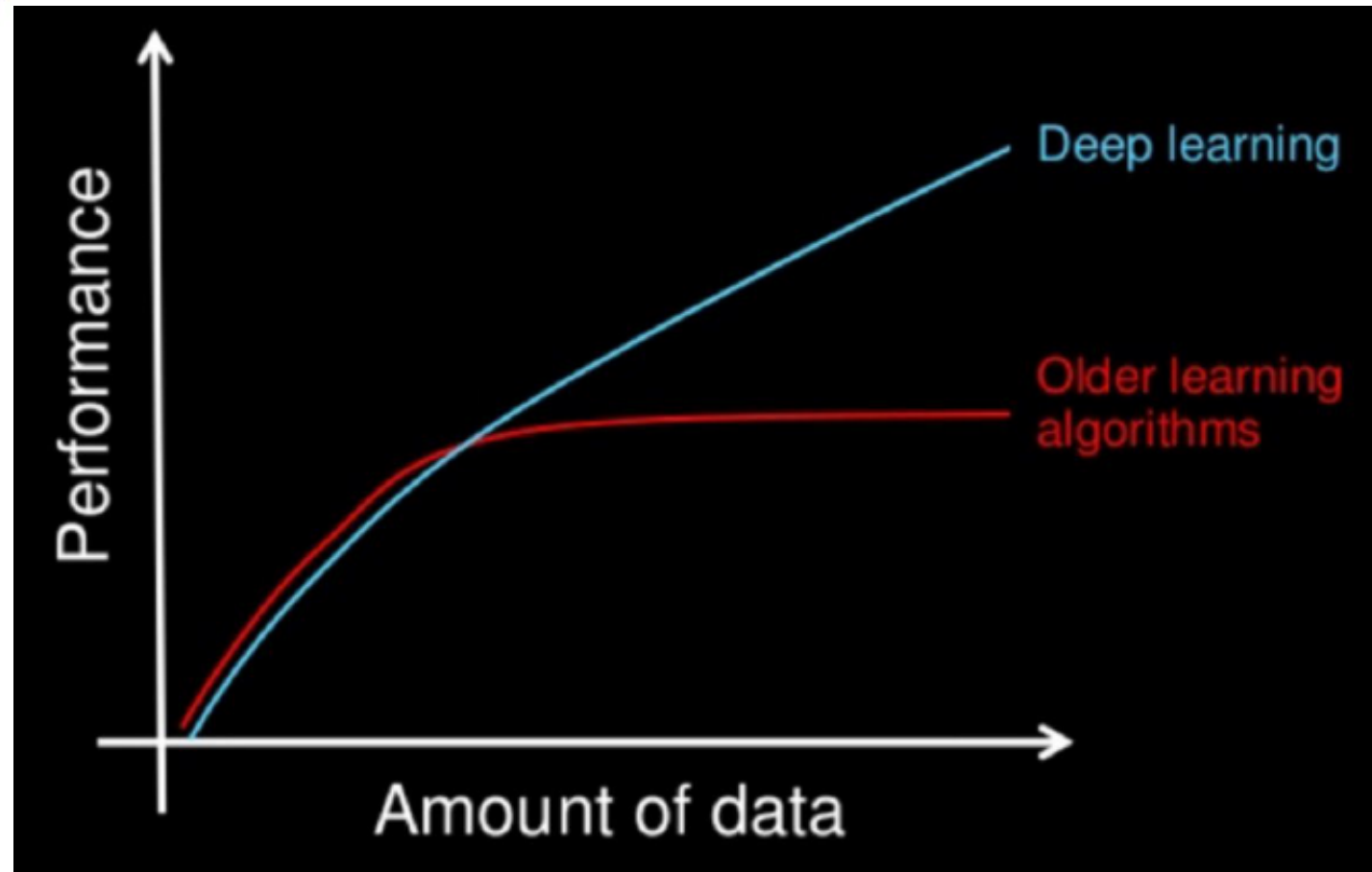


Deep Learning



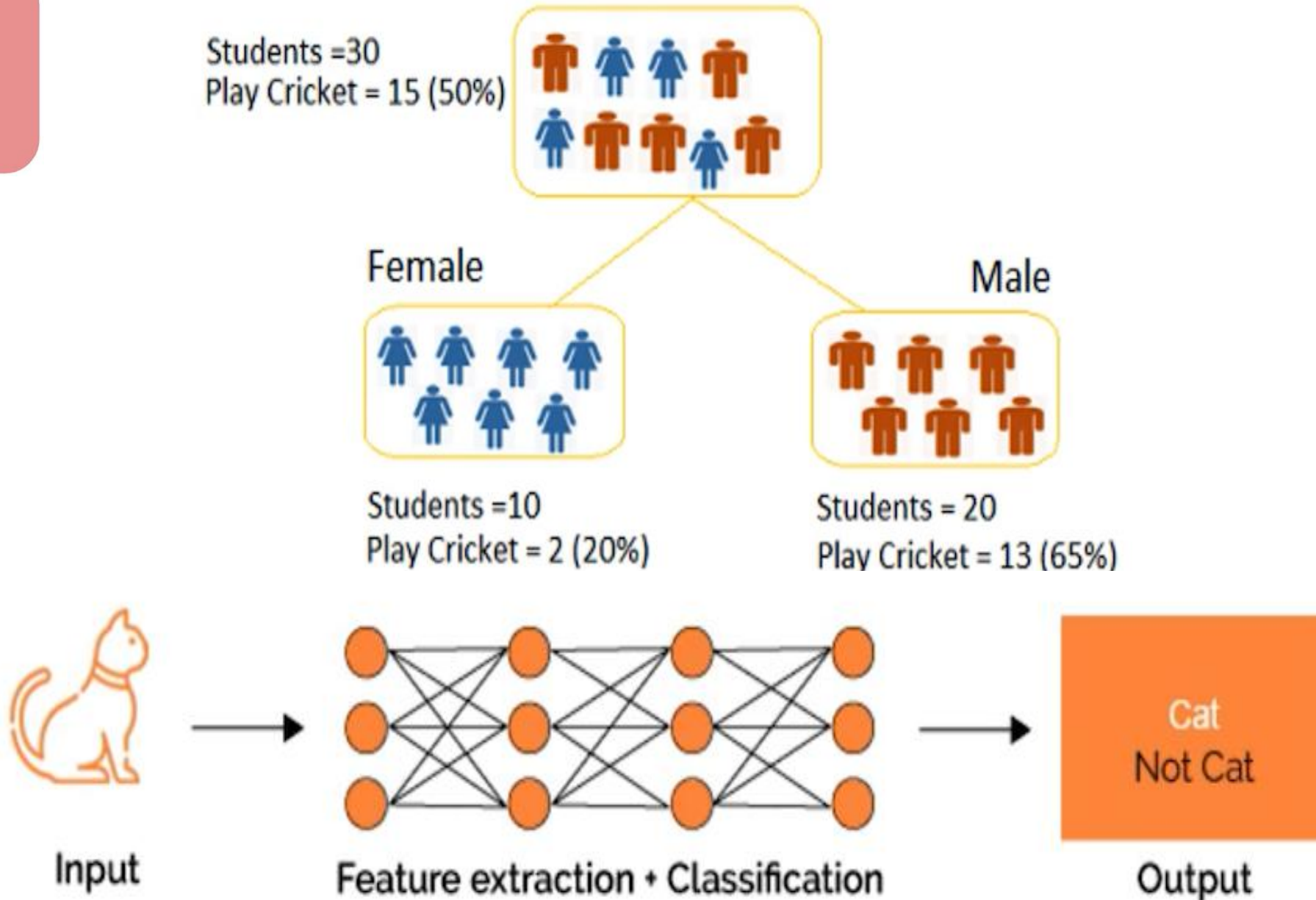
Machine Learning vs. Deep Learning

Performs well on
More Data



Machine Learning vs. Deep Learning

Less
Interpretable



Machine Learning vs. Deep Learning

Requires More
Computation
Power



CPU

- Small models
- Small datasets
- Useful for design space exploration



CPU

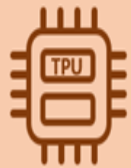


GPU

- Medium-to-large models, datasets
- Image, video processing
- Application on CUDA or OpenCL



GPU



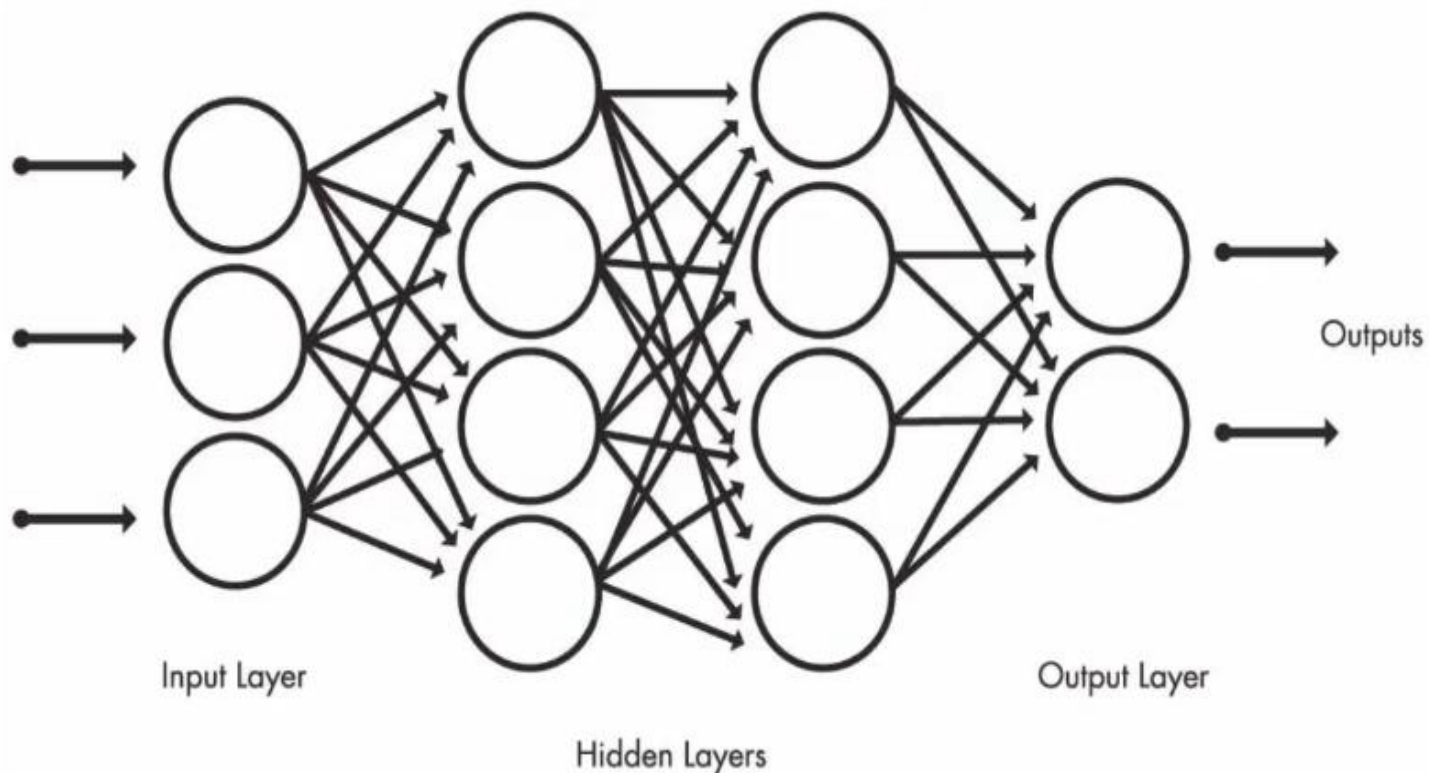
TPU

- Matrix computations
- Dense vector processing
- No custom TensorFlow operations



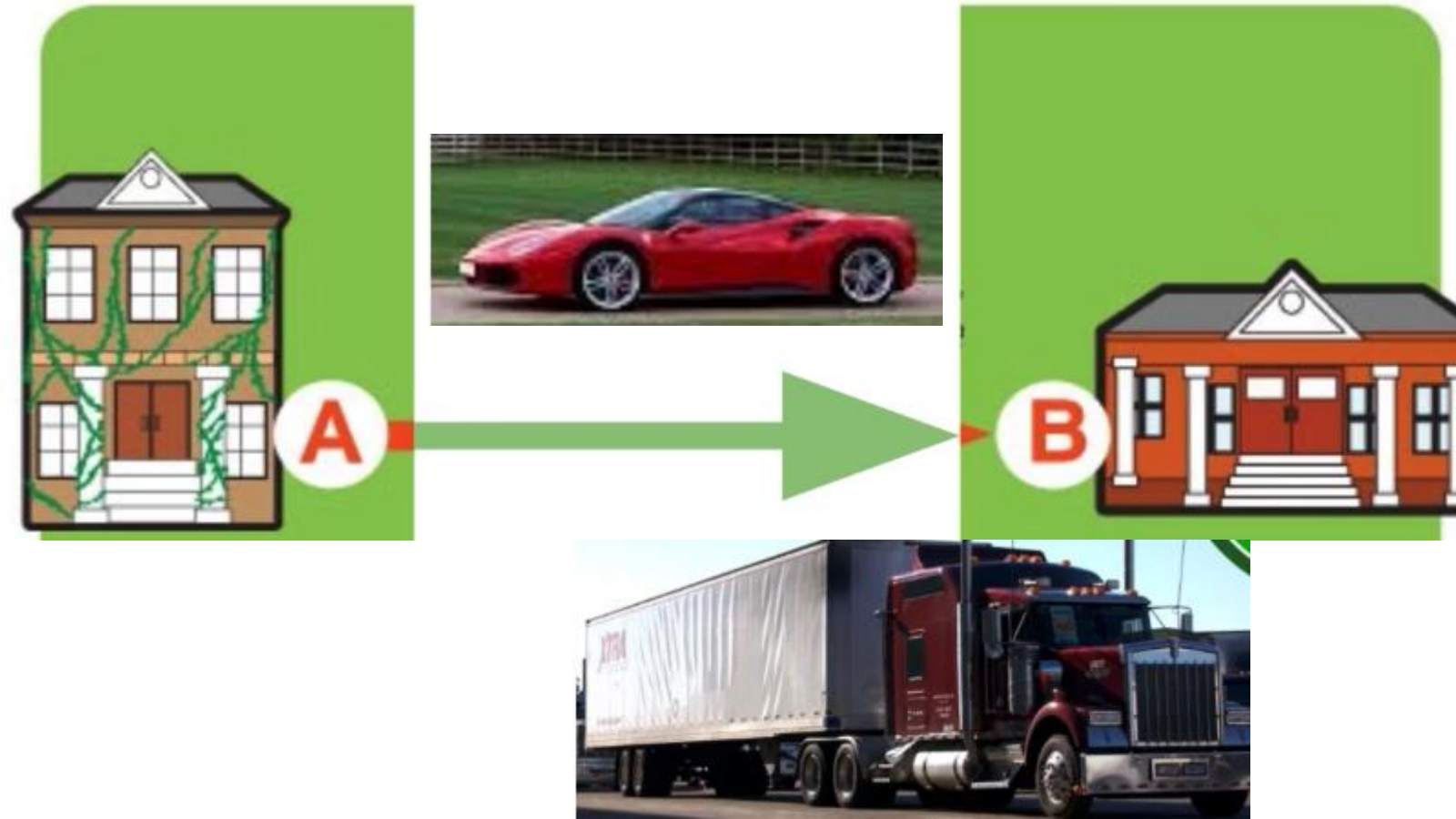
TPU

Hardware for Deep Learning



$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 7 \\ 9 \\ 11 \end{bmatrix} = \begin{bmatrix} 58 \\ 8 \\ 10 \\ 12 \end{bmatrix}$$
$$1 \cdot 7 + 2 \cdot 9 + 3 \cdot 11 = 58$$

Hardware for Deep Learning



CPU



GPU



TPU



Hardware for Deep Learning

CPU

Few complex cores

Focuses on doing one task as efficiently as possible

Used for general purpose tasks

Intel® Core™ i7-10510U Processor

4 cores



GPU

Hundreds of simpler cores

Focuses on doing various tasks parallelly

Used for graphics processing or matrix multiplication

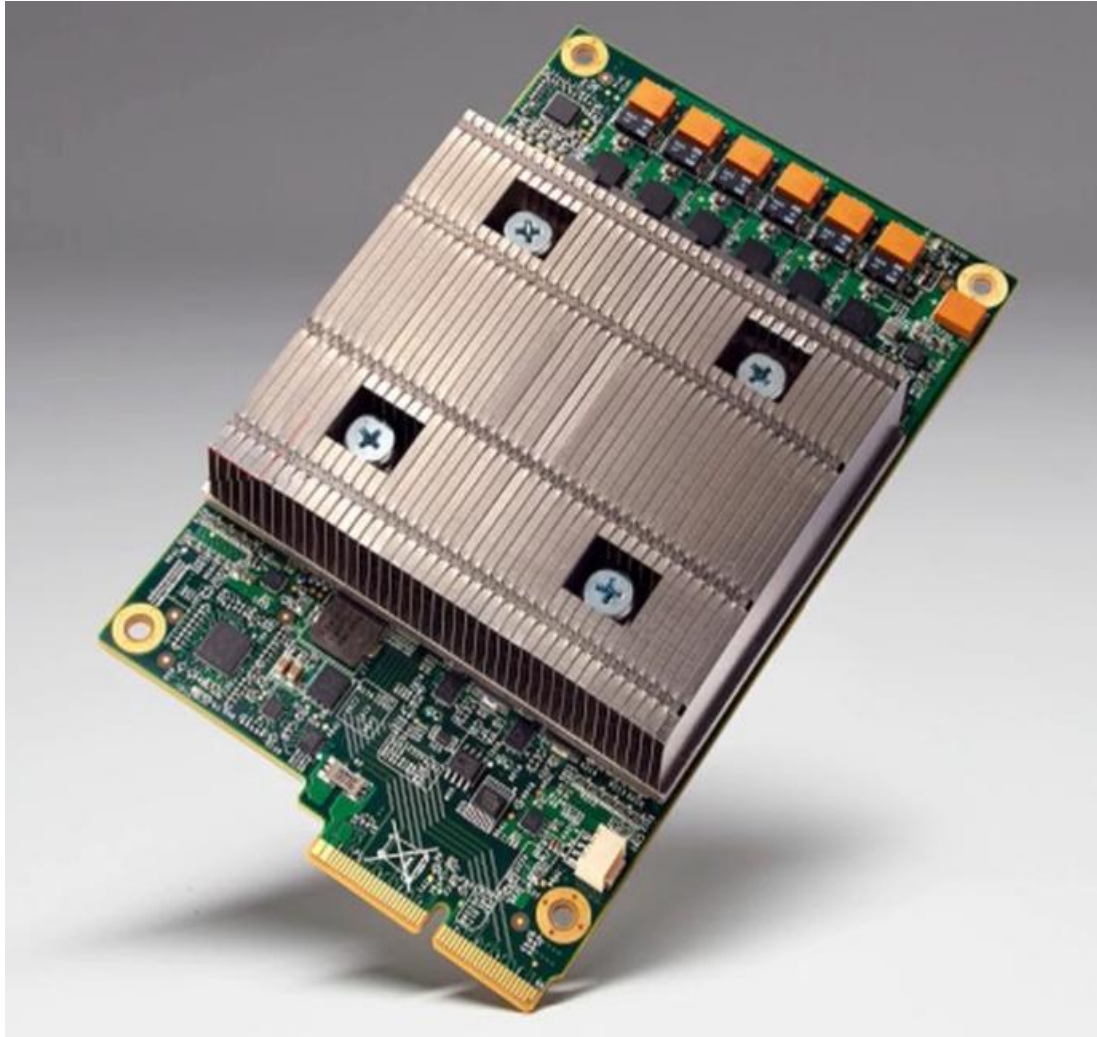
NVIDIA GTX 1080 Ti

3584 cores



Hardware for Deep Learning

Tensor Processing Units are an emerging hardware



Google Colaboratory



- Provides free GPU / TPU

Notebook settings

Runtime type

Python 3

Hardware accelerator

None

☐ Omit code cell output

None



GPU

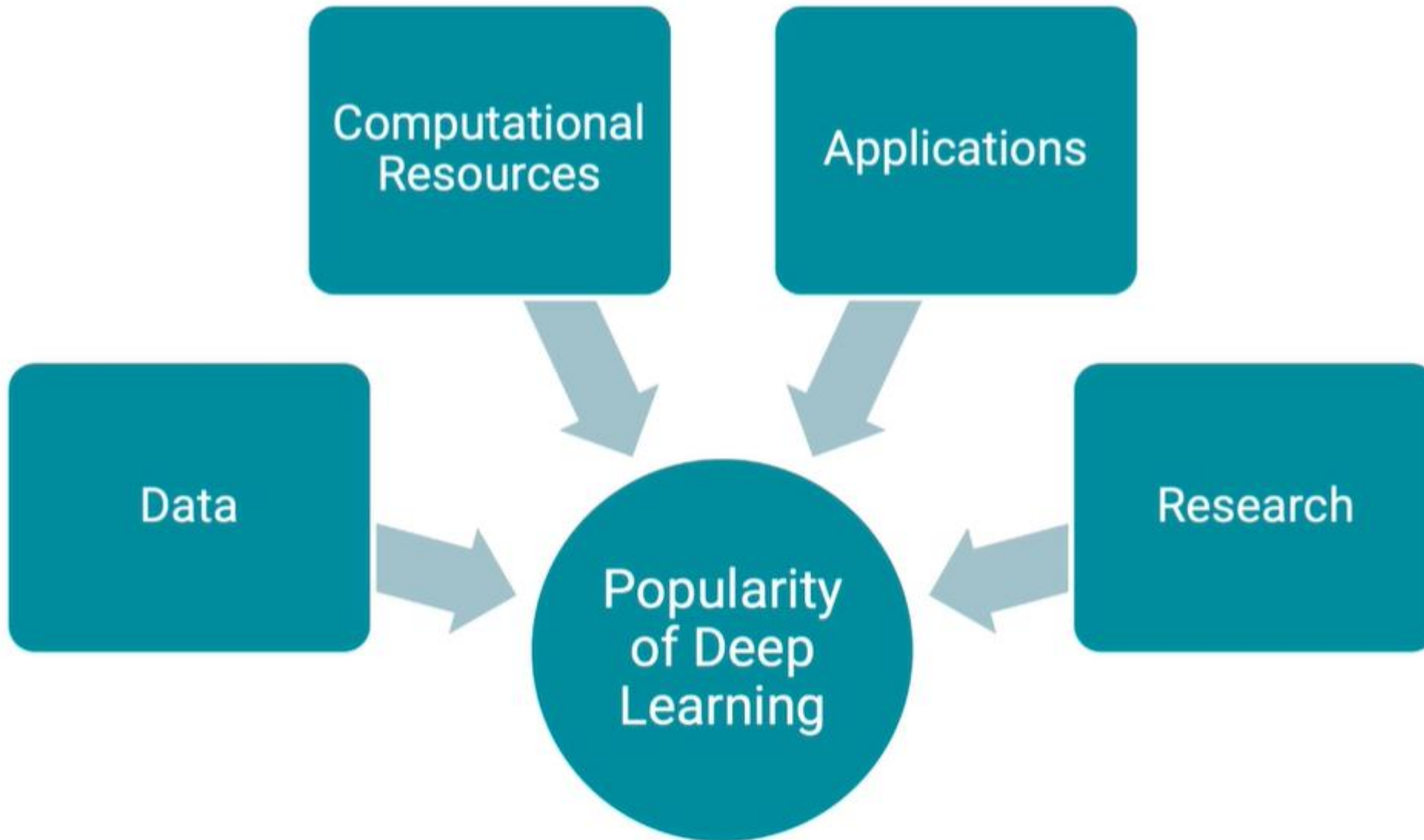
TPU

g this notebook

CANCEL

SAVE

Why Deep Learning has become so popular?



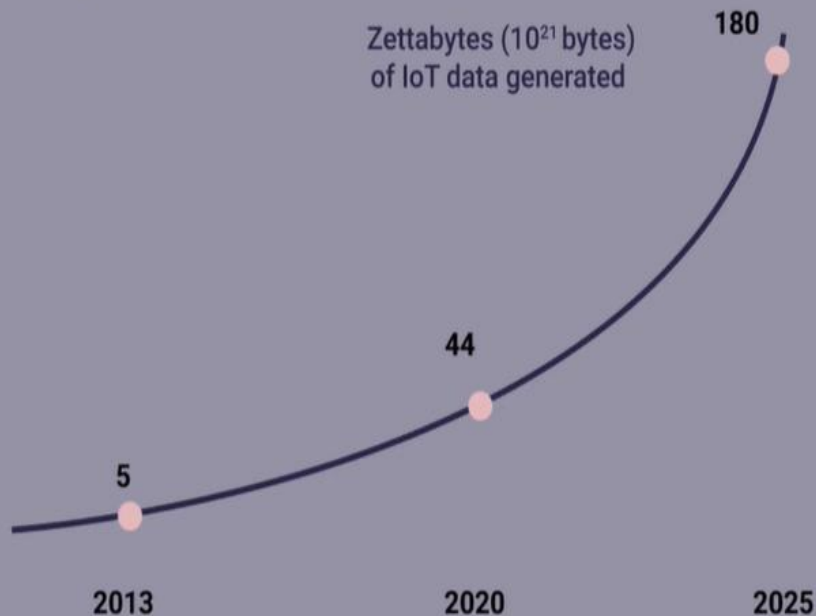
Why Deep Learning has become so popular?

Data

Source: <https://fortune.com/longform/ai-artificial-intelligence-deep-machine-learning/>

Data generation

Zettabytes (10^{21} bytes)
of IoT data generated



180 Zettabyte
of annual IoT
data generation
by 2025...

Data storage

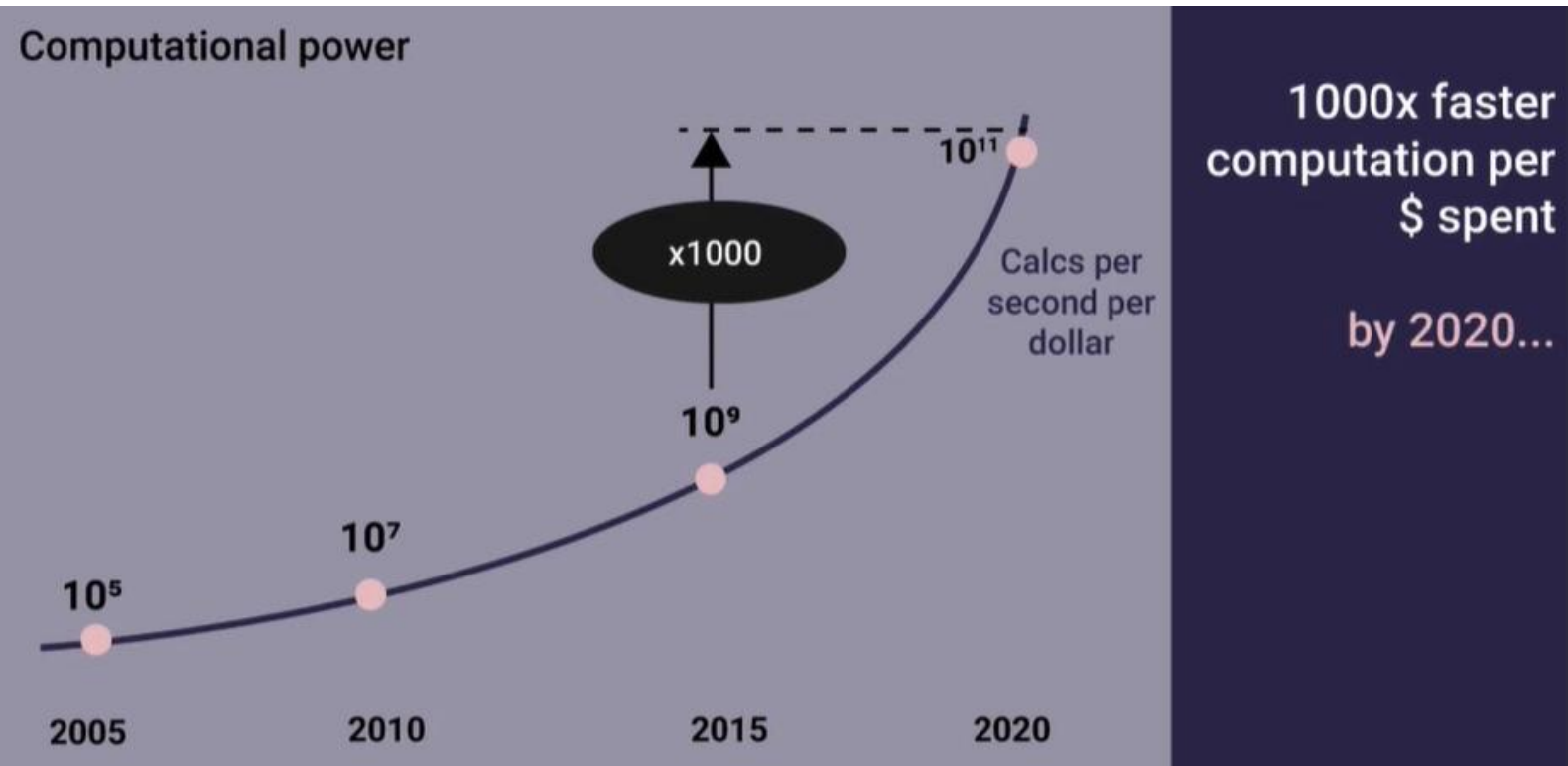
Cost per GB



Storage cost
perGB
approaching
\$0
by 2025...

Why Deep Learning has become so popular?

Computational Resources



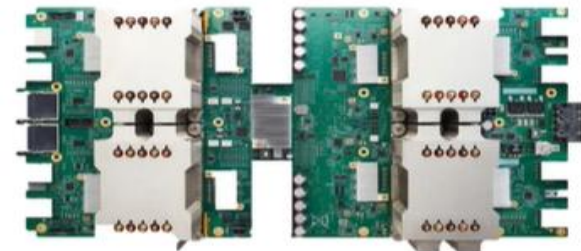
CPU



GPU



TPU



Why Deep Learning has become so popular?

Computational Resources

- Google Colaboratory (Both GPU and TPU)
- Kaggle Kernels
- Paperspace
- Vast.AI
- Oracle Cloud
- Amazon Web Services (AWS)
- Microsoft Azure
- Google Cloud Platform (GCP)



Why Deep Learning has become so popular?

Applications

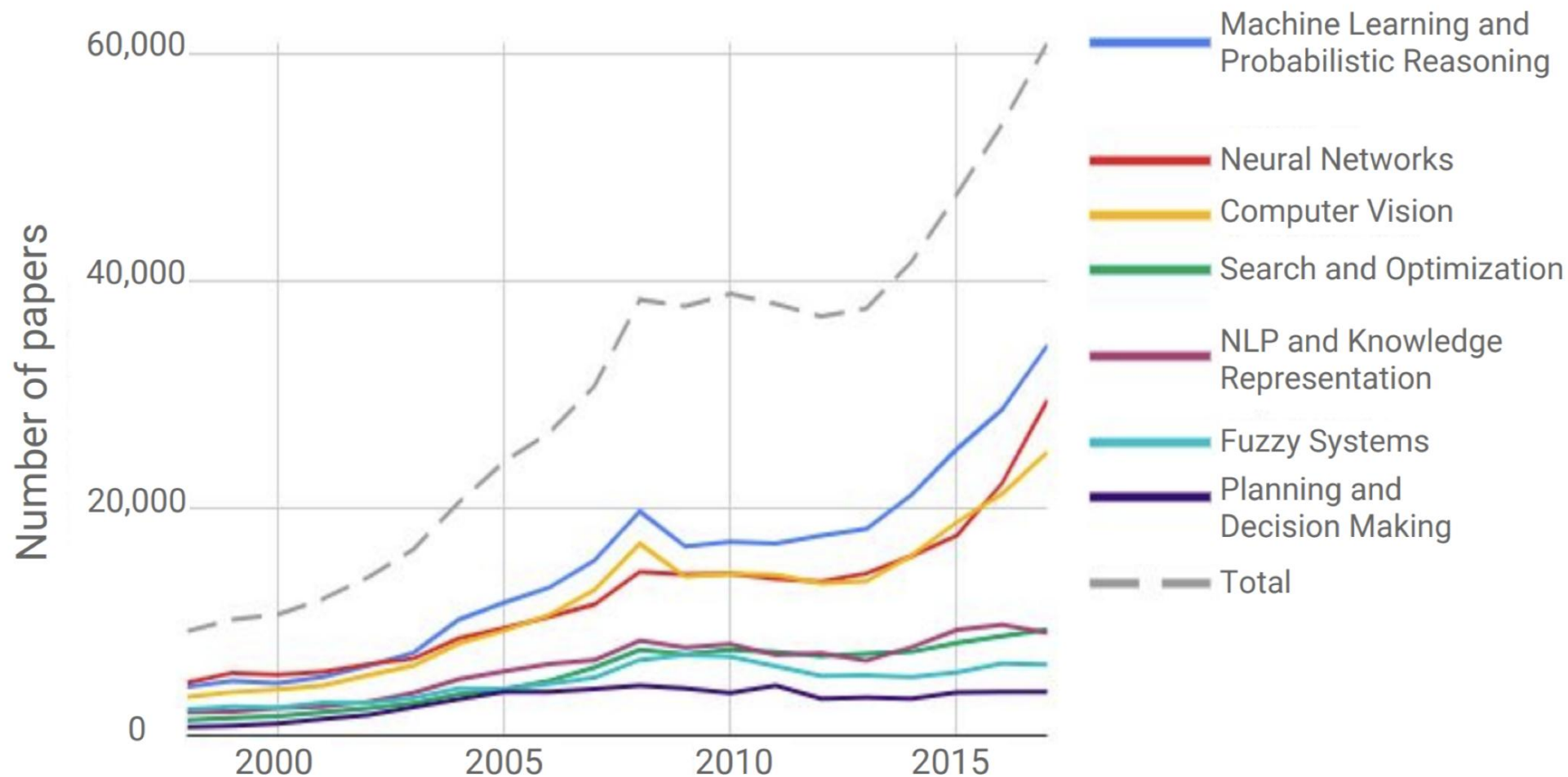


Why Deep Learning has become so popular?

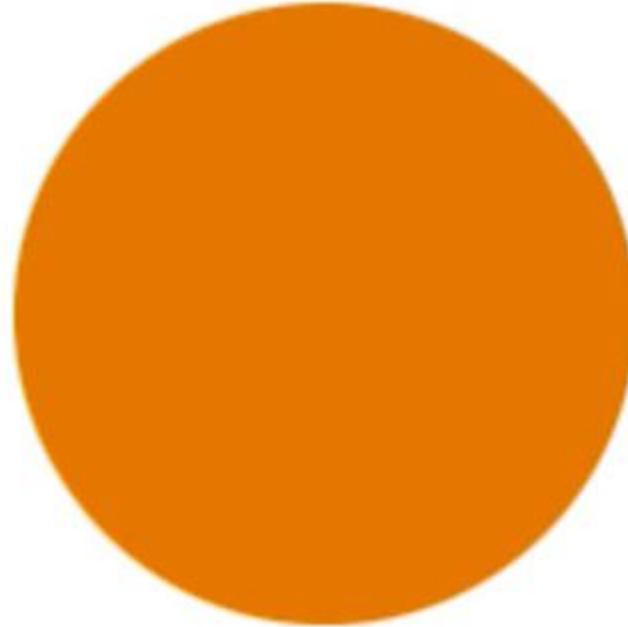
Research

Number of AI papers on Scopus by subcategory (1998–2017)

Source: Elsevier



How Deep Learning Model Works?



1. Four Lines
2. They are connected, Closed, perpendicular to each other
3. All lines are equal in length

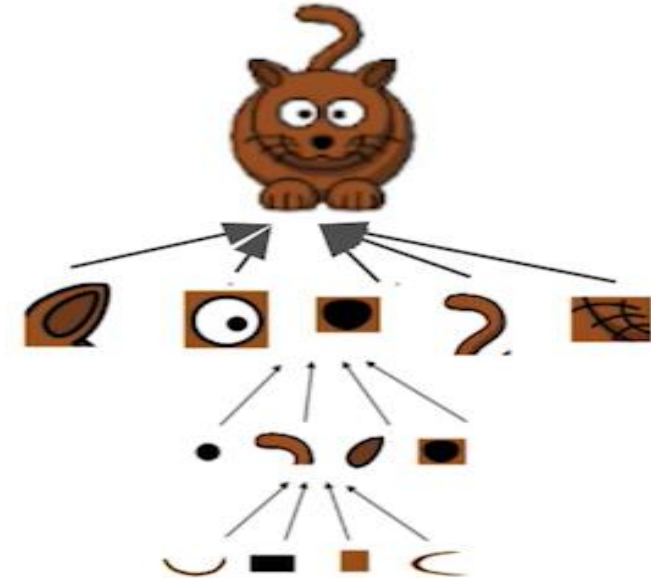
REMEMBER THIS:

A Deep Learning Model breaks a complex concept into a hierarchy of simpler concepts.

How Deep Learning Model Works?



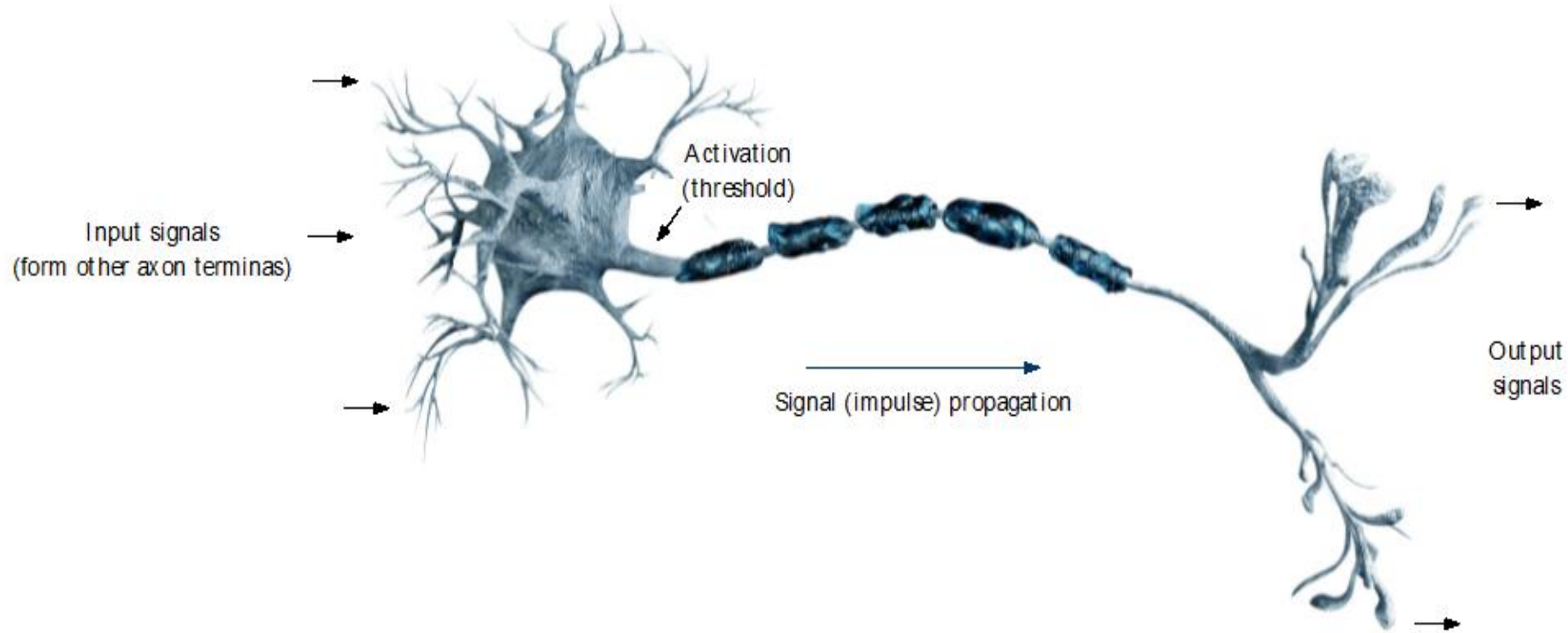
1. Identifies what are the edges
2. Builds on this hierarchically to find combination of shapes and edges
3. Identifies which of these features are responsible for Cat Vs. Dog classification



A deep learning model works **hierarchically** — first seeing edges, then shapes, and finally **learning which combinations represent a cat or a dog** — all based on training data.

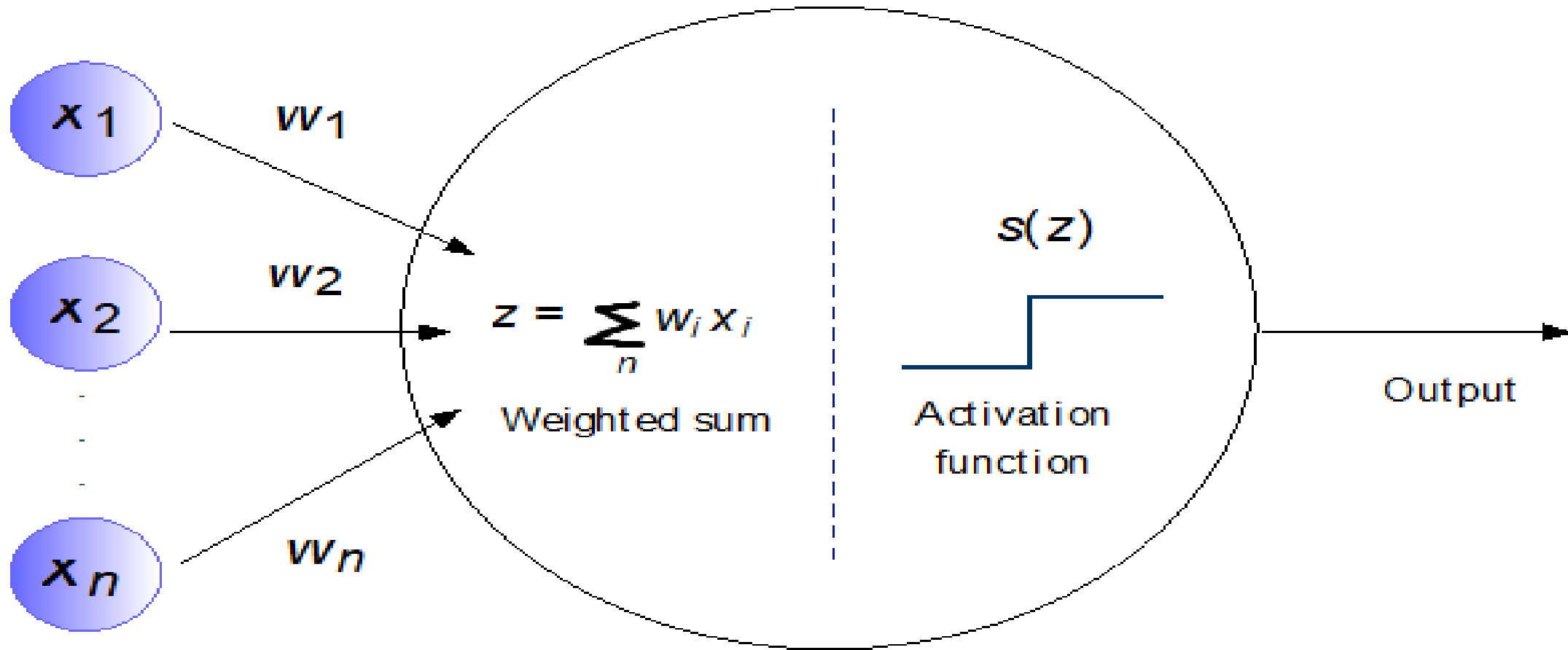
Introduction to Artificial Neural Network

Biological Neurons



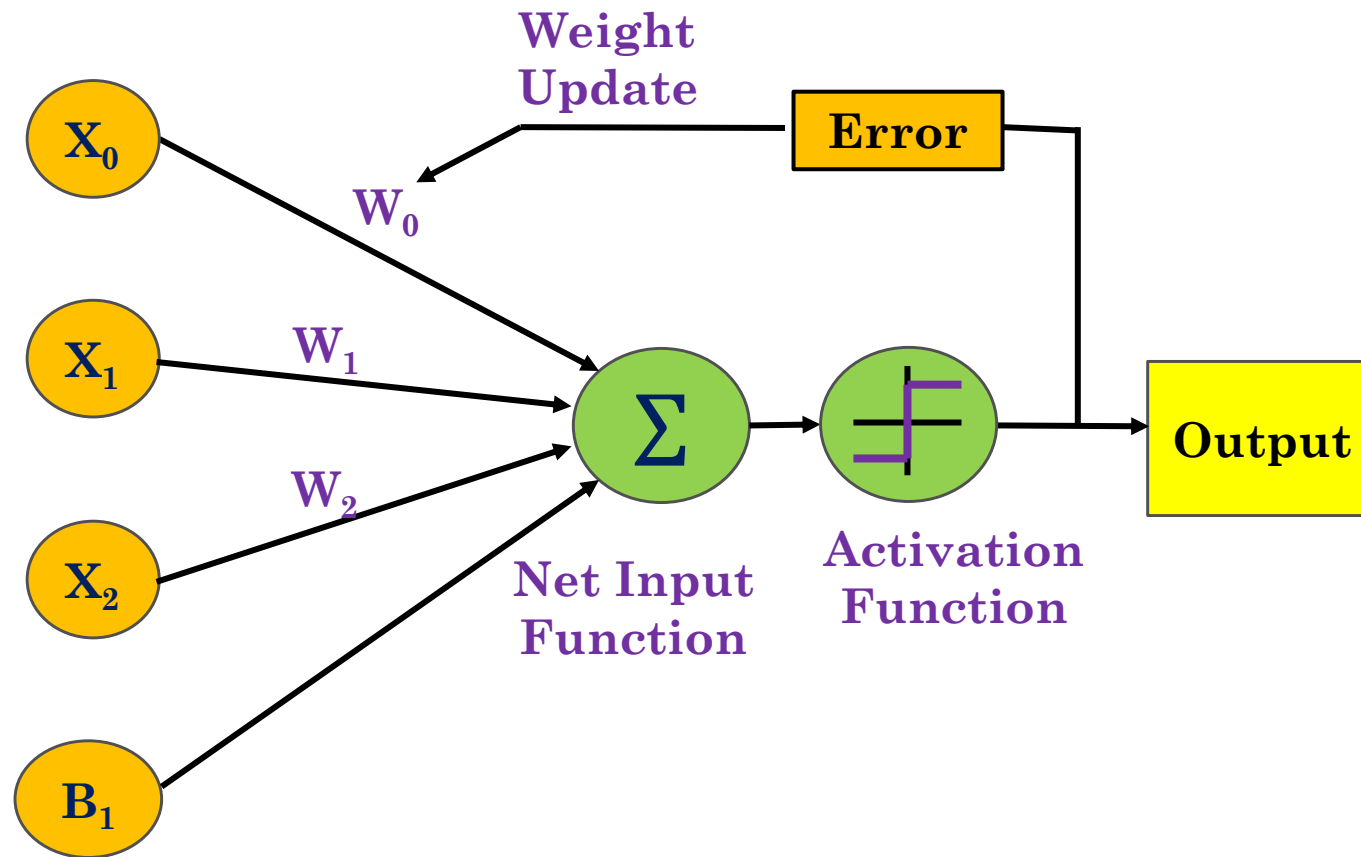
1. Inputs arrive from other neurons / nodes.
2. Receiving neuron / node processes the inputs $\rightarrow$ weighted sum.
3. Receiving neuron / node applies an activation function (sigmoid).
4. Output of activation function passed to subsequent neurons / nodes.

Artificial Neuron: Overview



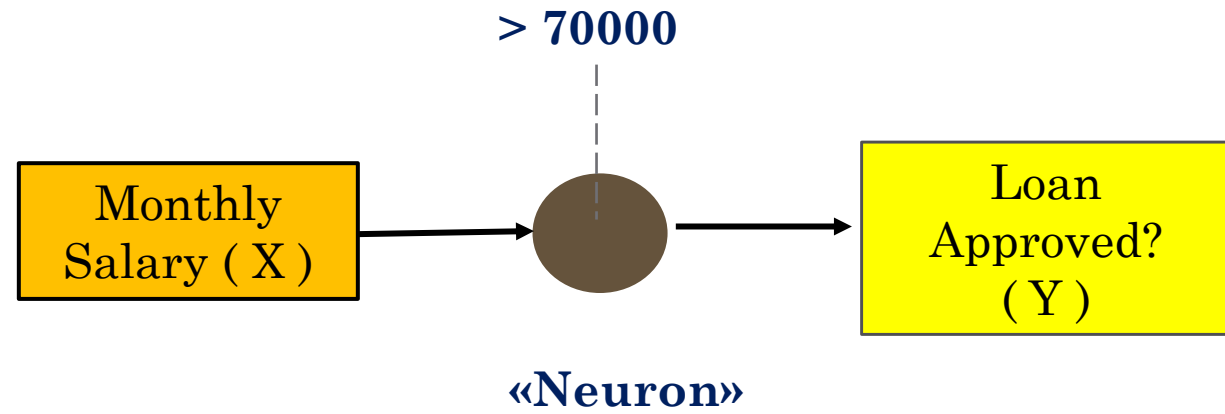
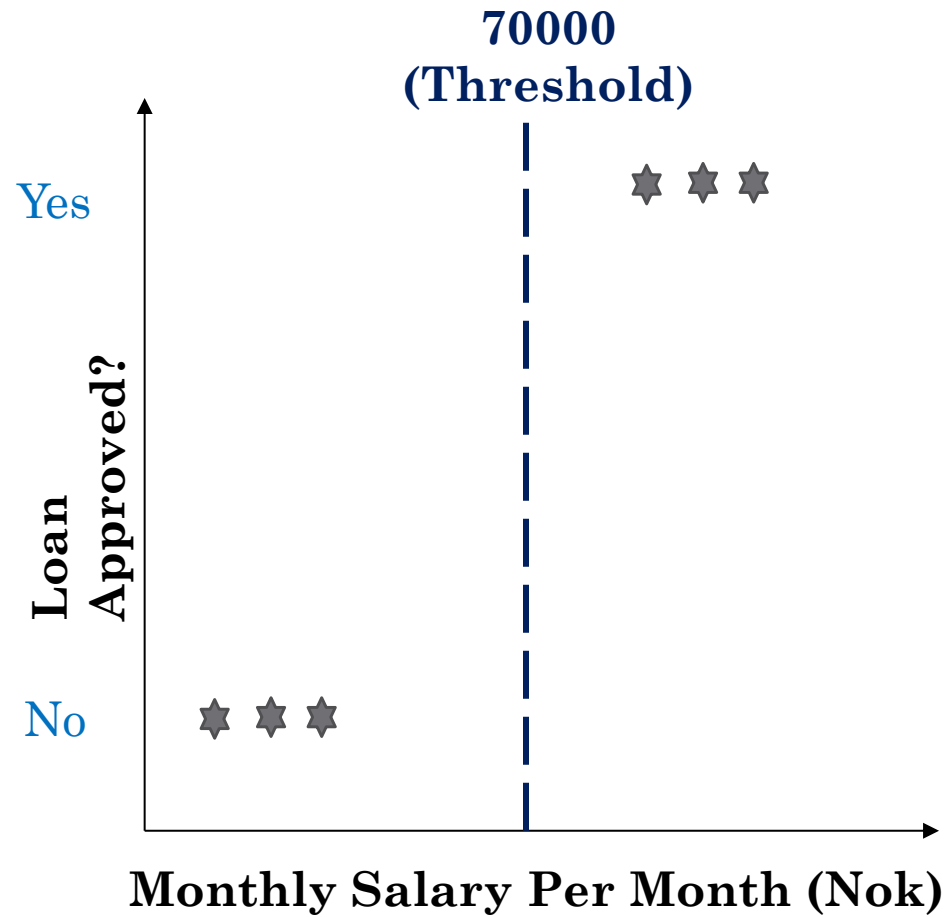
- Origins: Algorithms that try to mimic the brain
- Were very widely used in the 1980s and 90s: popularity diminished in the late 1990s.
- Recent resurgence: State-of-the-art technique for many applications.

Artificial Neuron: Overview

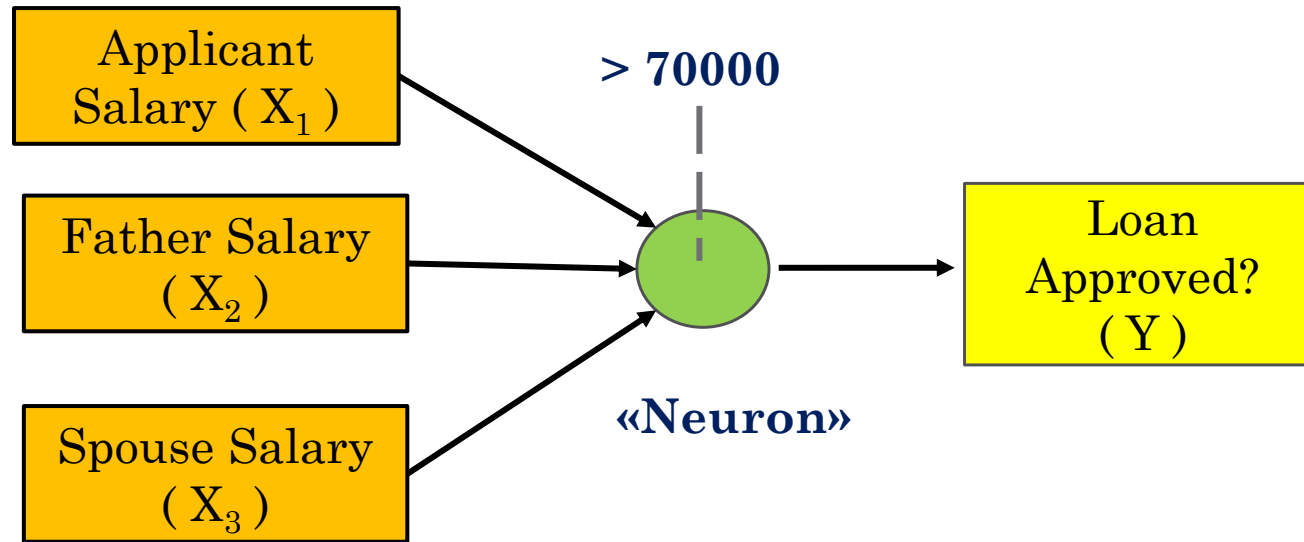


BREAK

Intuition Behind Perceptron



Understanding Perceptron



Total Income > Threshold?

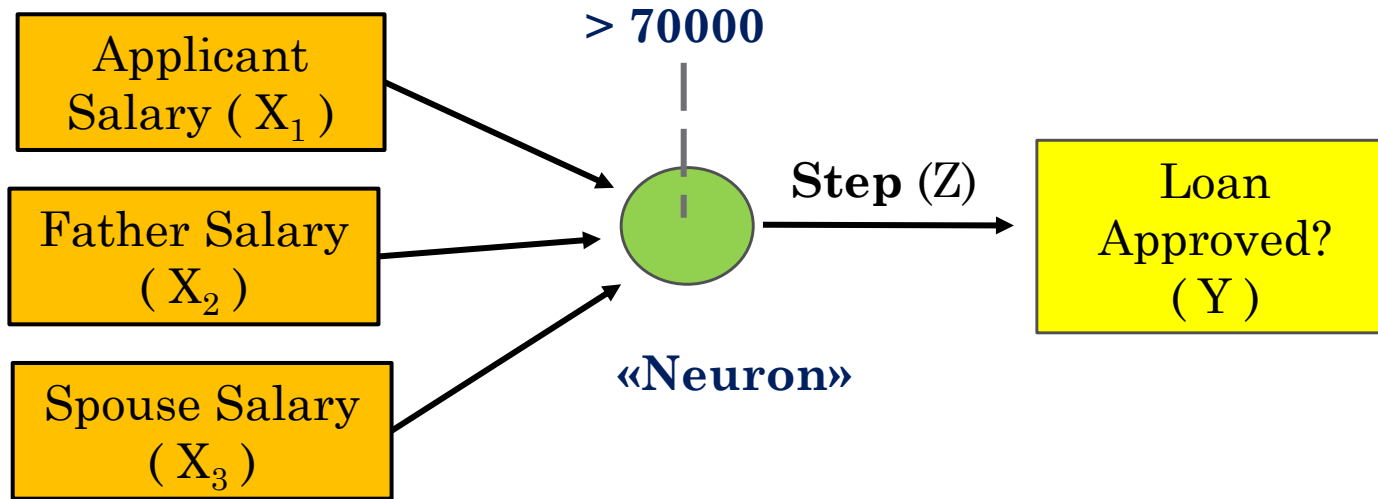
$$X_1 + X_2 + X_3 > \text{Threshold}$$

$$X_1 + X_2 + X_3 - \text{Threshold} > 0$$

$$X_1 + X_2 + X_3 + \text{Bias} > 0$$

Where minus Threshold is equal to Bias

Understanding Perceptron



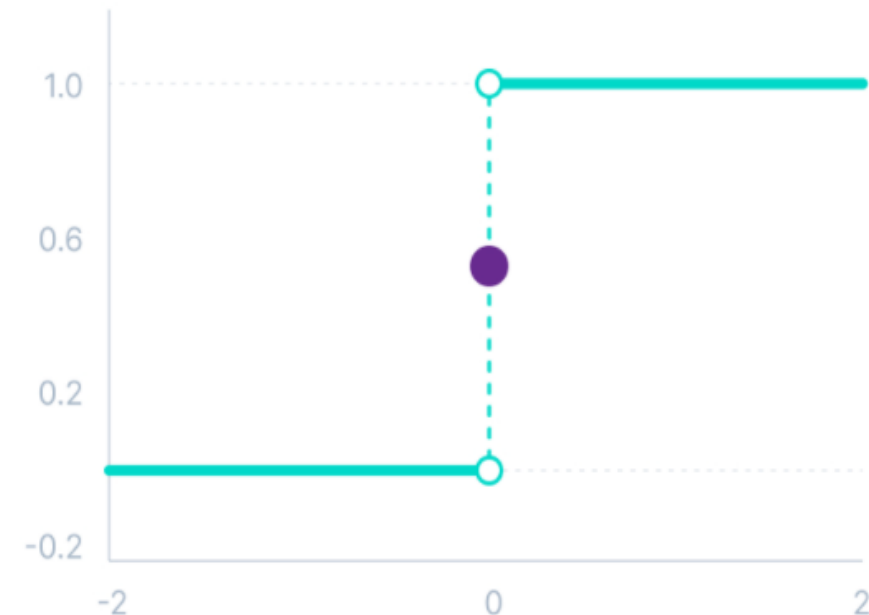
If $X_1 + X_2 + X_3 + \text{Bias} > 0$ then output should be 1

If $X_1 + X_2 + X_3 + \text{Bias} < 0$ then output should be 0

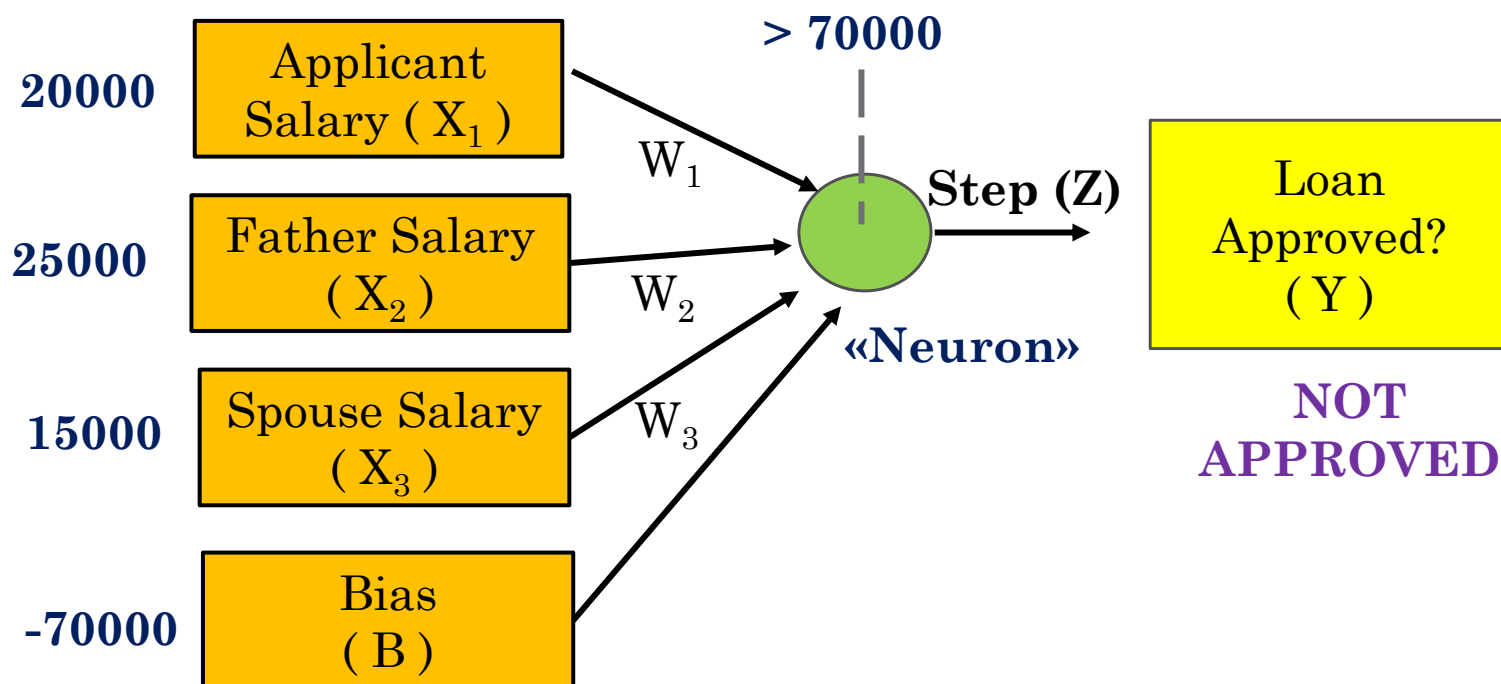
Now substituting $X_1 + X_2 + X_3 + \text{Bias} = Z$

$$\text{Output} = \begin{cases} 1, Z > 0 \\ 0, Z \leq 0 \end{cases}$$

Binary Step Function



Understanding Perceptron



$X_1 * W_1$	$X_2 * W_2$	$X_3 * W_3$	Sum of Inputs	Z (Sum of Inputs + Bias)	Step (Z)
20000*1	25000*1	15000*1	60000	-10000	0

Quiz

As discussed these are the tasks performed by a perceptron model. Kindly arrange them in the order they are executed

- a. Take salary as input
- b. If the condition is true, then output is a "Yes"
- c. Check if its at least 50,000 or not

A

a -> b -> c

B

b -> a -> c



a -> c -> b

D

a -> b -> c

Quiz

In a neural network, what does the bias term represent?

A It is the sum of the inputs



It is the negative of threshold value

C It is the output of the model

What is a perceptron?

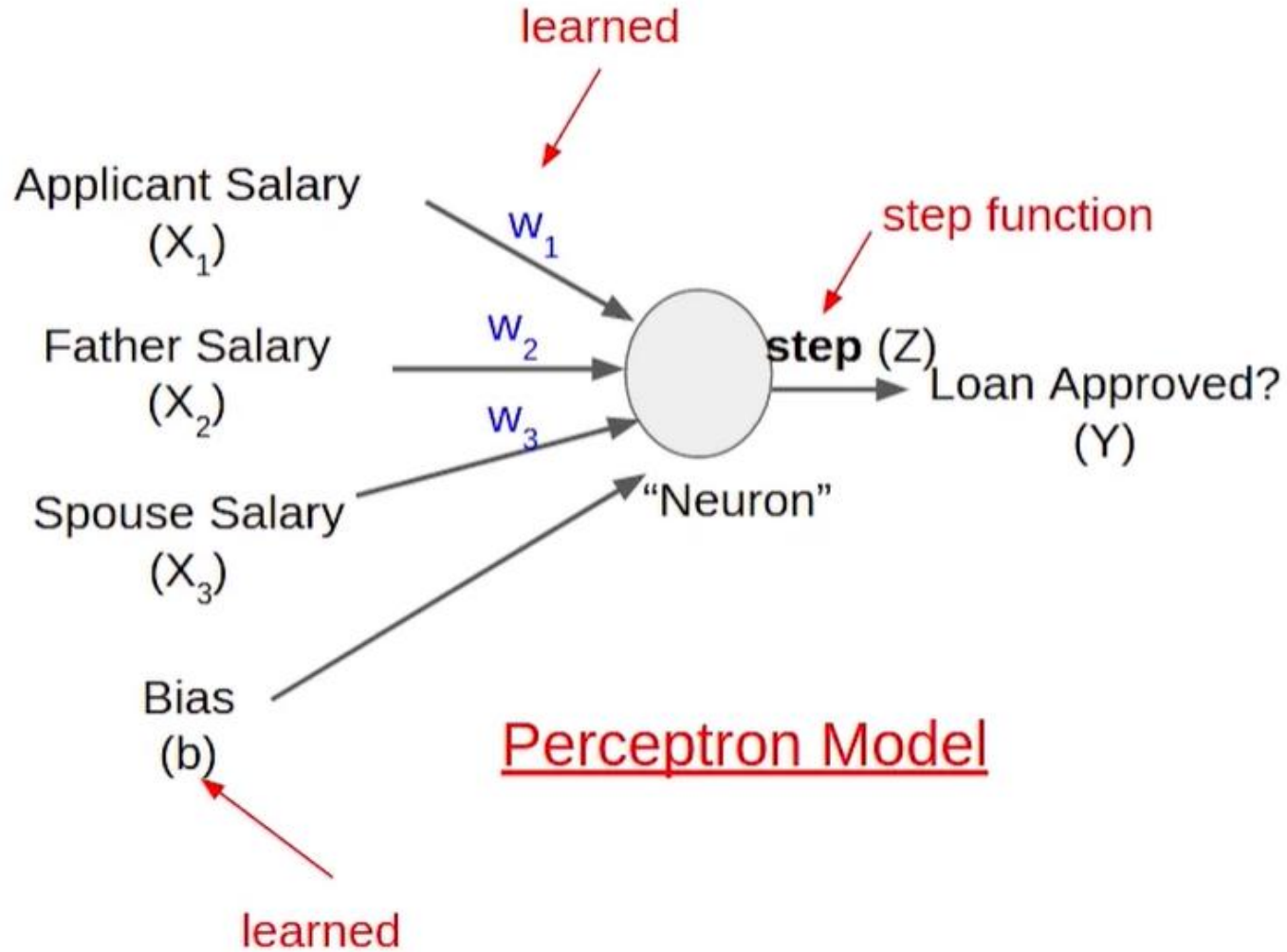
A Each neuron is known as a perceptron.

B A neuron with no activation function is a perceptron



A neuron with step function as the activation function is a perceptron

Weights in Perceptron

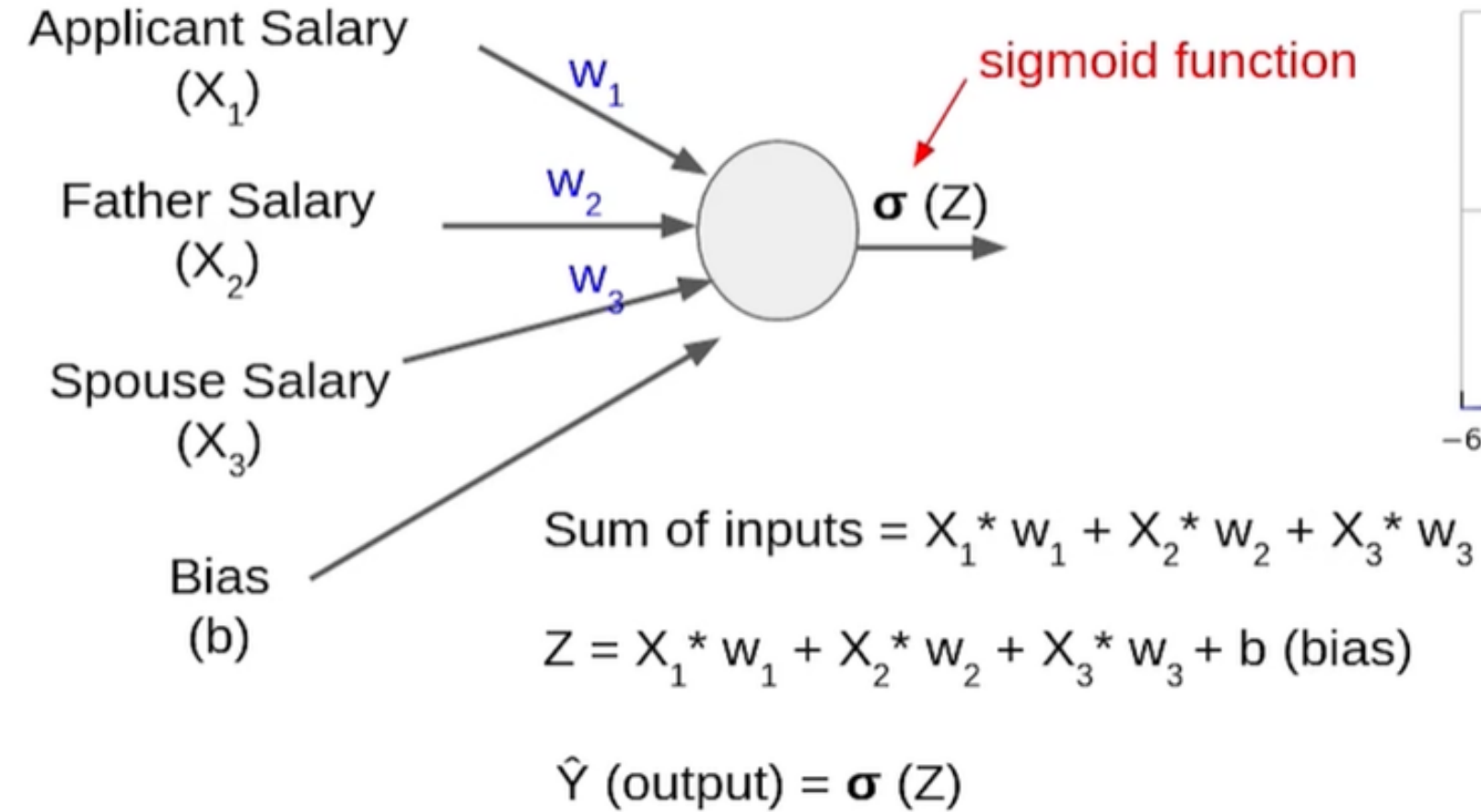


$$\text{Sum of inputs} = X_1 * w_1 + X_2 * w_2 + X_3 * w_3$$

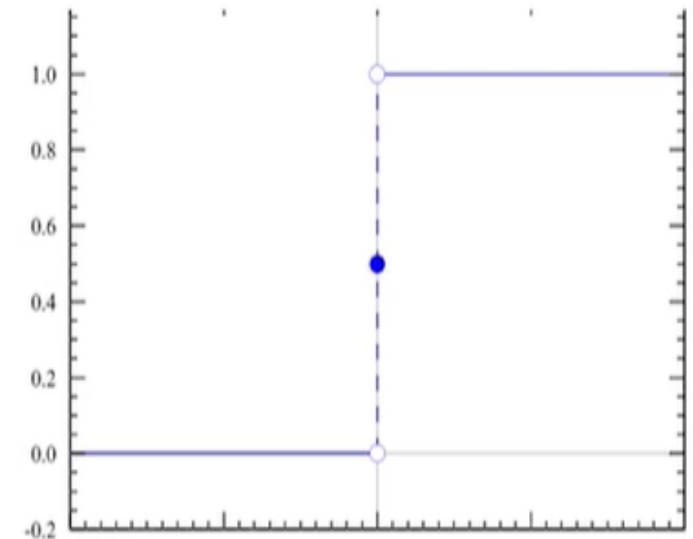
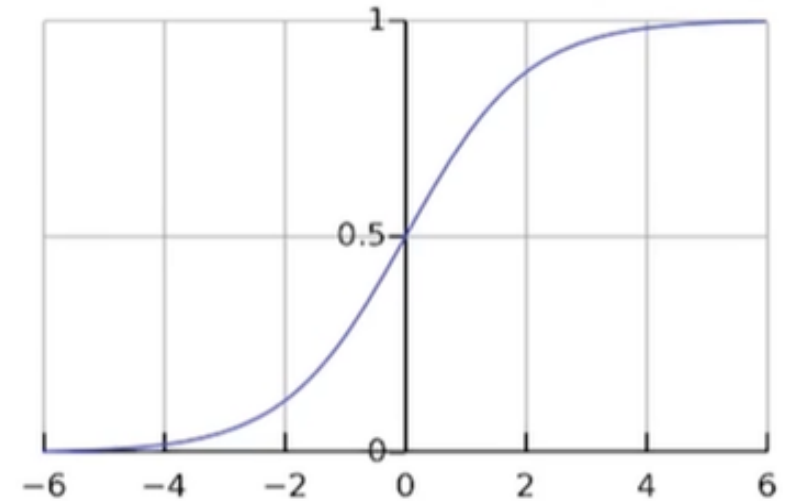
$$Z = X_1 * w_1 + X_2 * w_2 + X_3 * w_3 + b \text{ (bias)}$$

$$\hat{Y} \text{ (output)} = \text{step} (Z)$$

Sigmoid Neuron



$$\text{Output} = \begin{cases} 1, & Z > 0 \\ 0, & Z \leq 0 \end{cases}$$



Quiz

What is the difference between a perceptron and logistic regression?

Choose only ONE best answer.



Activation function in case of perceptron is step function and in case of logistic regression it is sigmoid

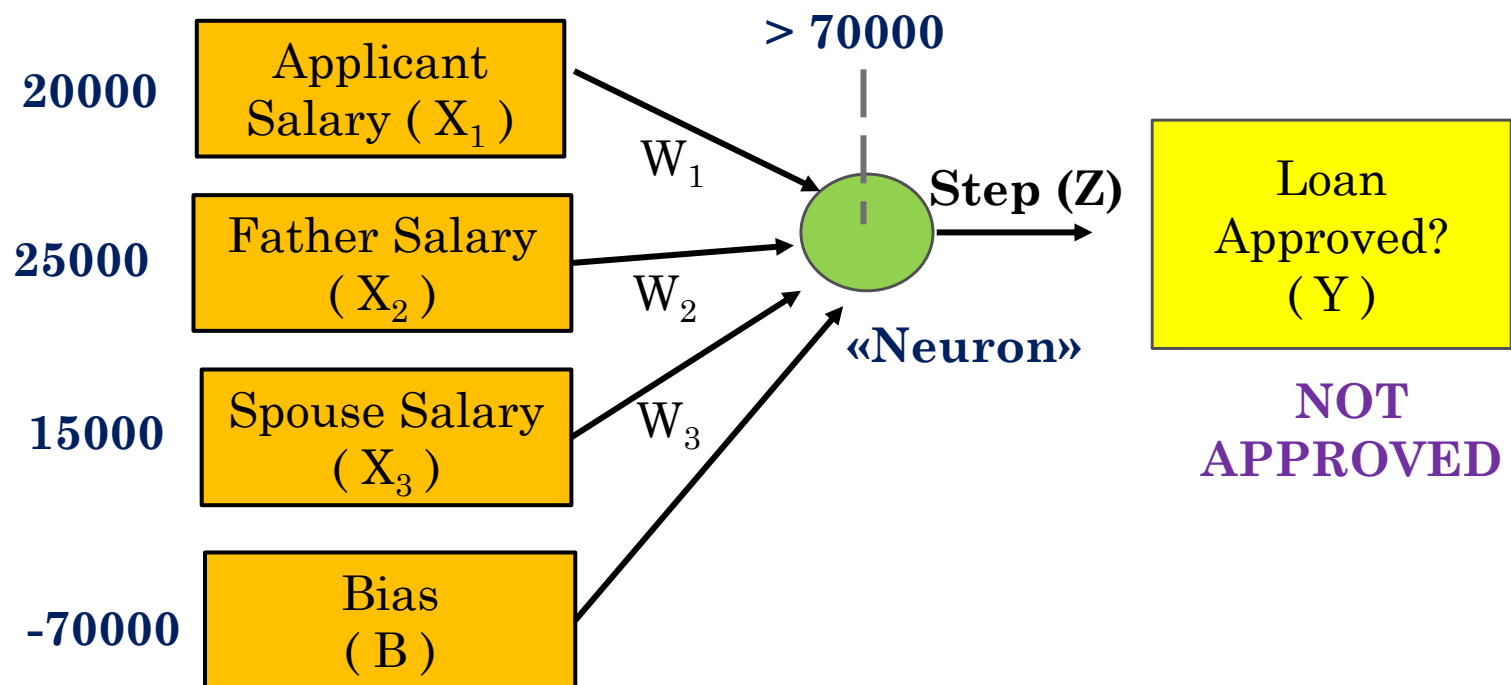
B

Activation function in case of perceptron is sigmoid and in case of logistic regression it is step function

C

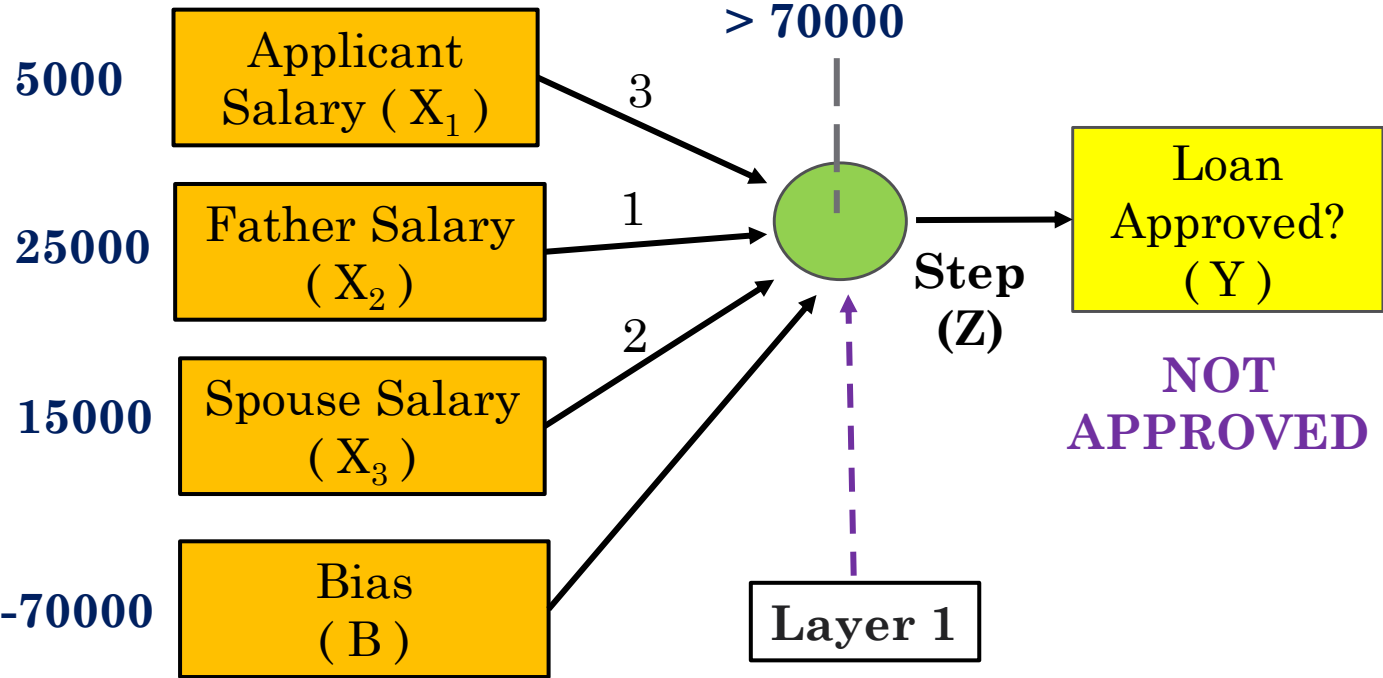
Both perceptron and logistic regression are same

Single Layer Perceptron



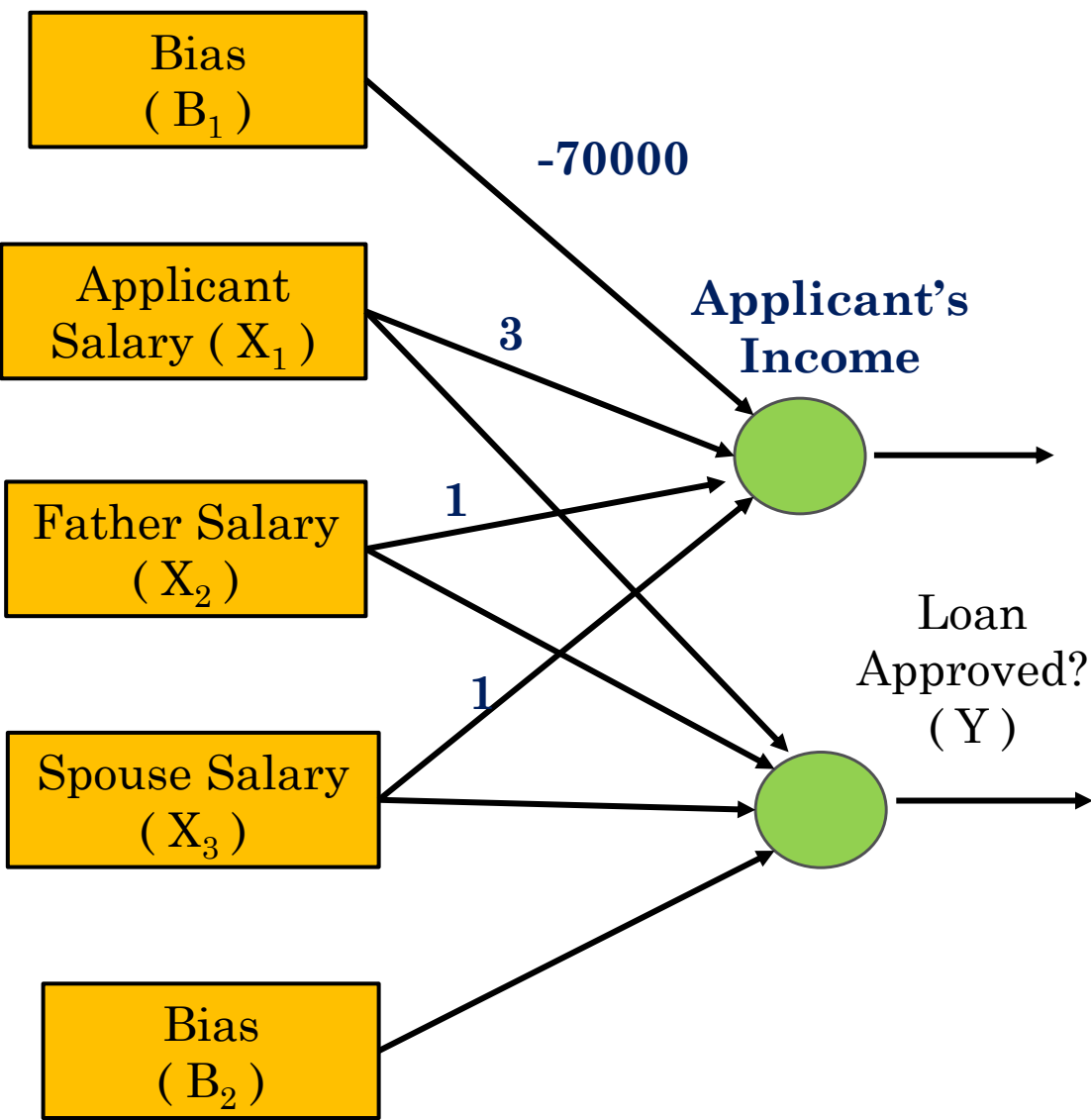
$X_1 * W_1$	$X_2 * W_2$	$X_3 * W_3$	Sum of Inputs	Z (Sum of Inputs + Bias)	Step (Z)
$20000 * 1$	$25000 * 1$	$15000 * 1$	60000	-10000	0

Single Layer Perceptron

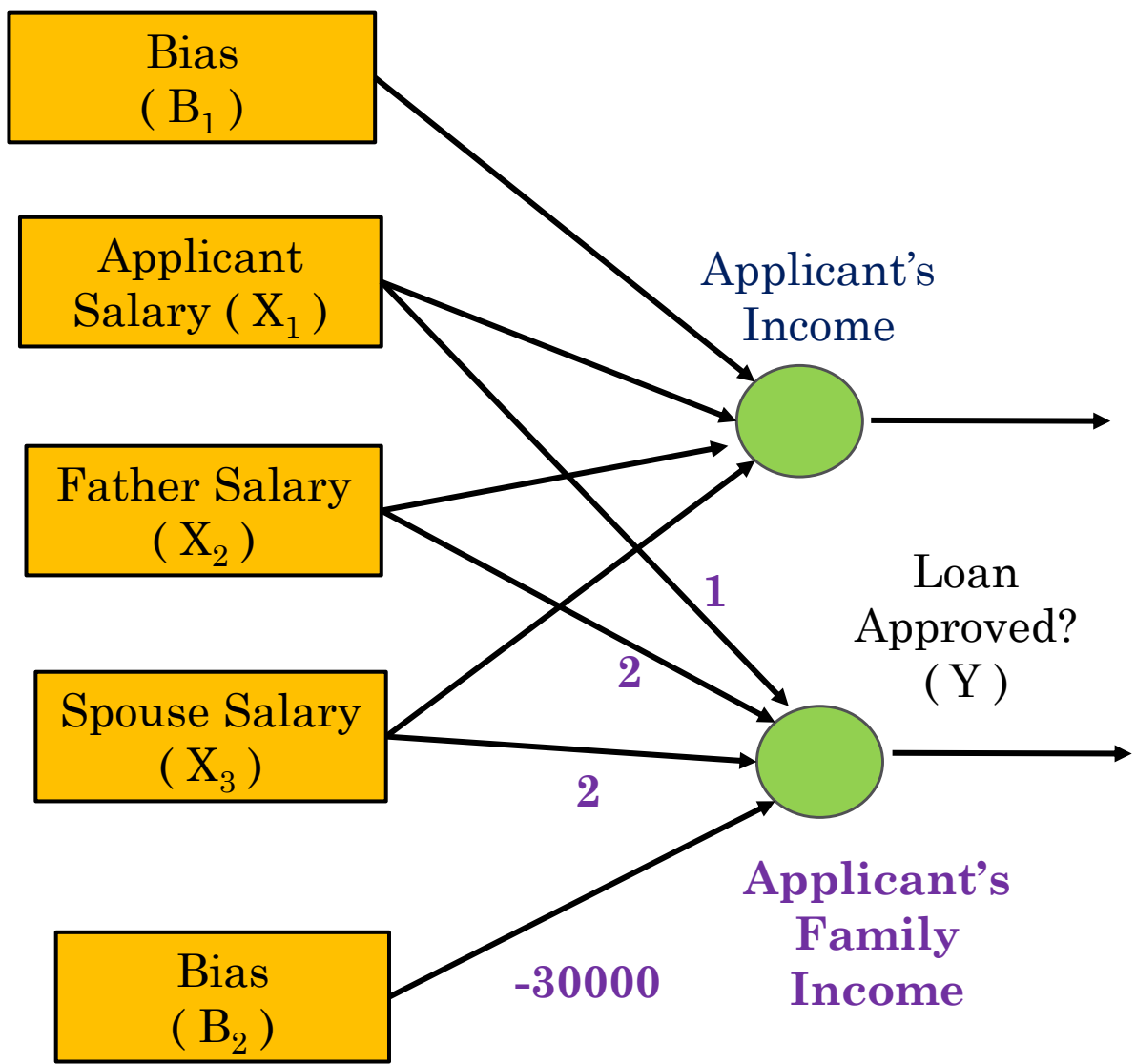


$X_1 * W_1$	$X_2 * W_2$	$X_3 * W_3$	Sum of Inputs	Z (Sum Inputs + Bias)	Step (Z)
5000*3	25000*1	15000*2	70000	0	0

Single Layer Perceptron



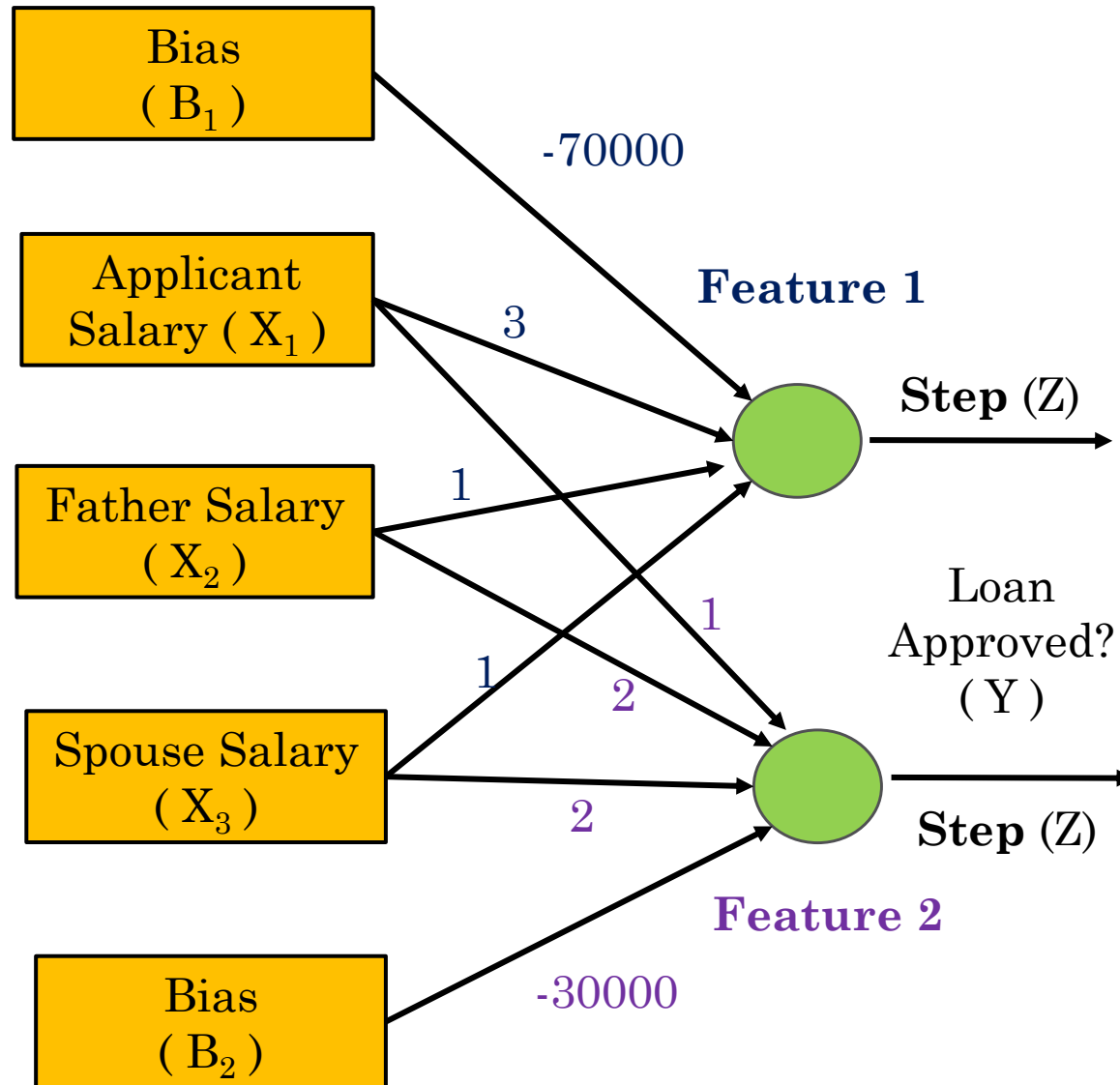
Focus on Applicant's Income



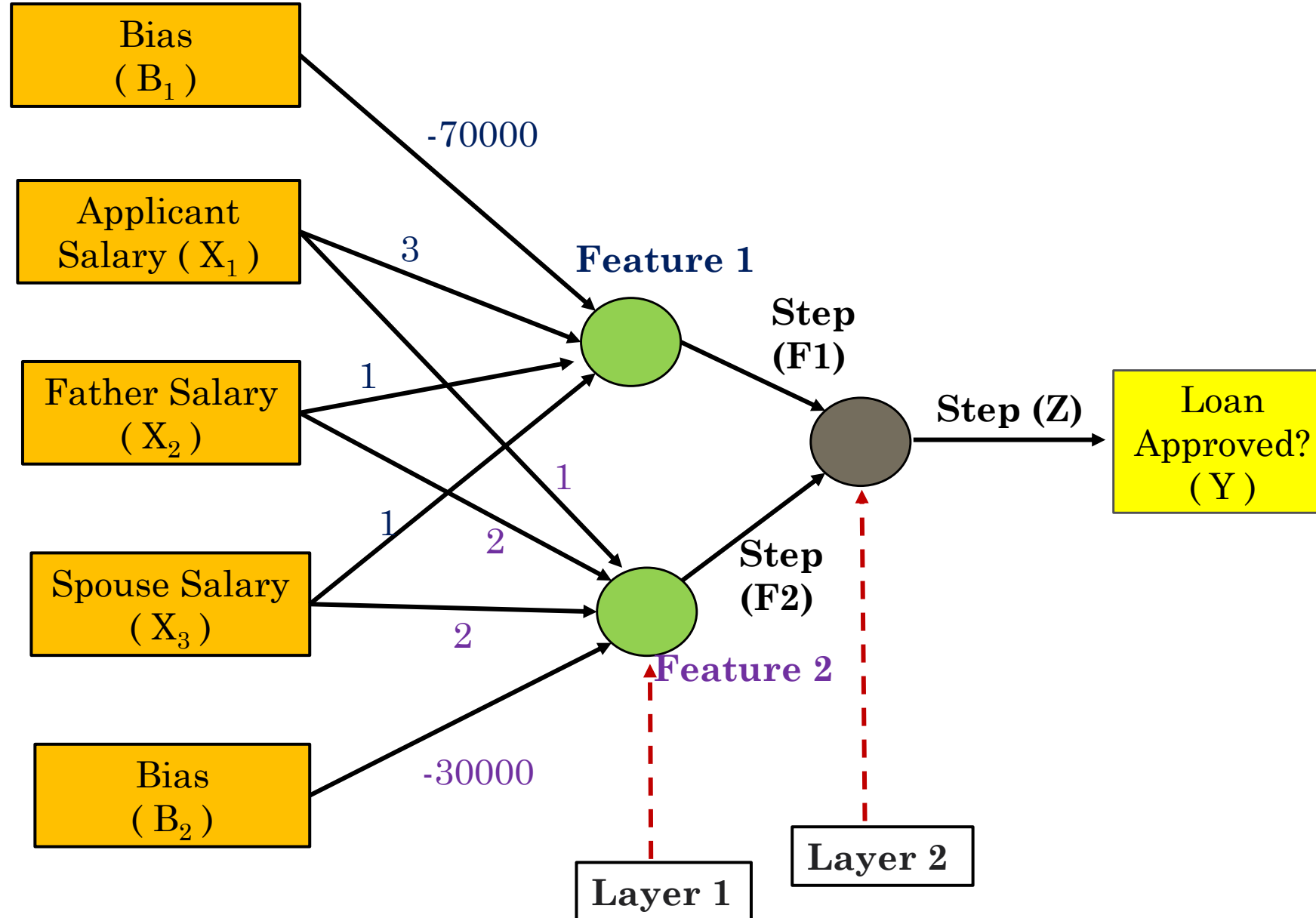
Focus on Applicant's Family Income

Introduction to Neural Network: Multilayer Perceptron

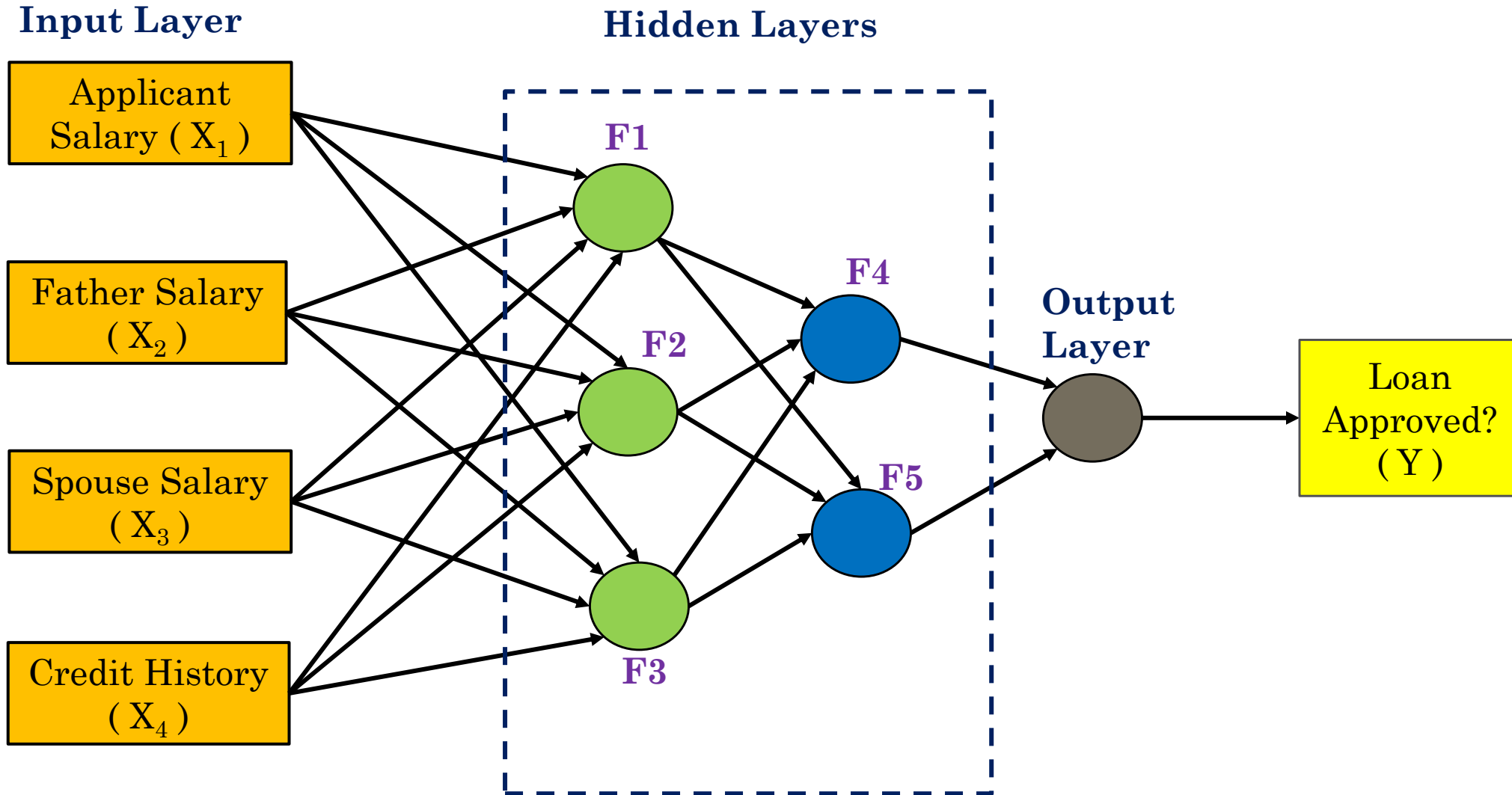
Multi Layer Perceptron



Multi Layer Perceptron

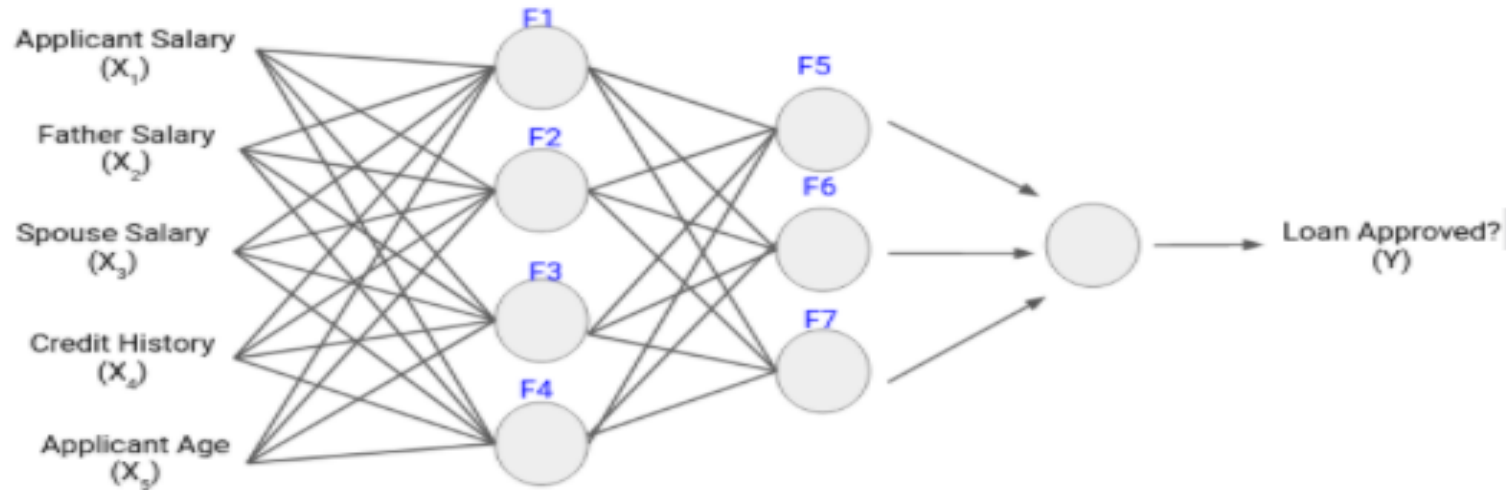


Multi Layer Perceptron



Quiz

How many Hidden layers does this model have?



A

One

✗

Two

C

Three

D

Four

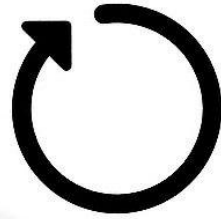
BREAK

Forward and Backward Propagation

Epoch vs. Batch vs. Iteration

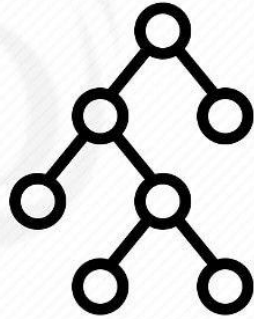
Epoch :

An Epoch represent one iteration over the entire dataset.



Batch :

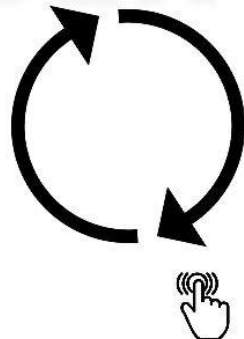
We cannot pass the entire dataset into the Neural Network at once. So, we divide the dataset into number of batches.



Iteration = 10

Iteration :

If we have 1000 images as Data and a batch size of 20, then an Epoch should run $1000/20 = 50$ iteration.



Total data : 1,000

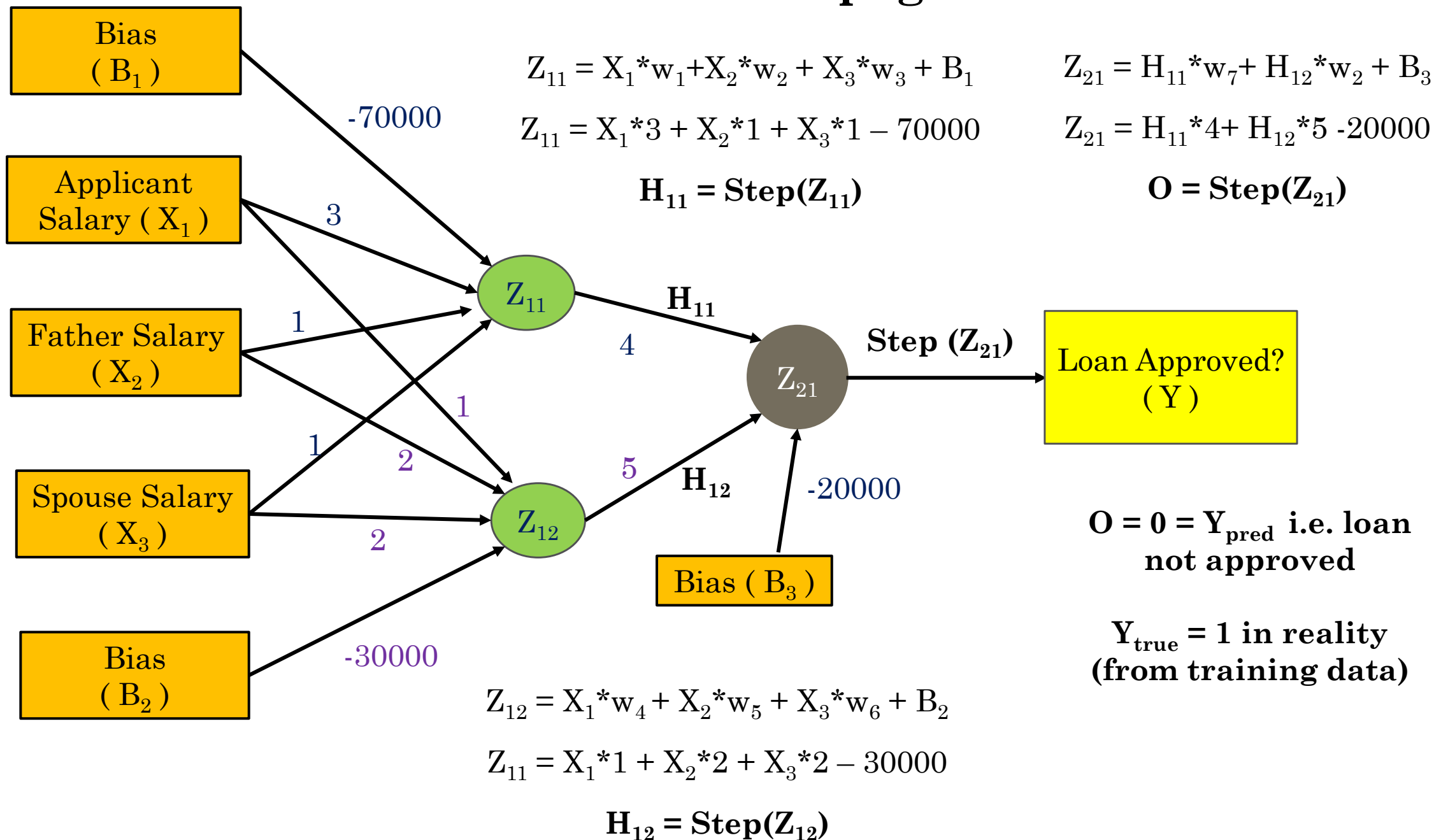
100
100
100
100
100
100
100
100
100
100
100

} Batch size = 100



+1 epoch !

Forward Propagation

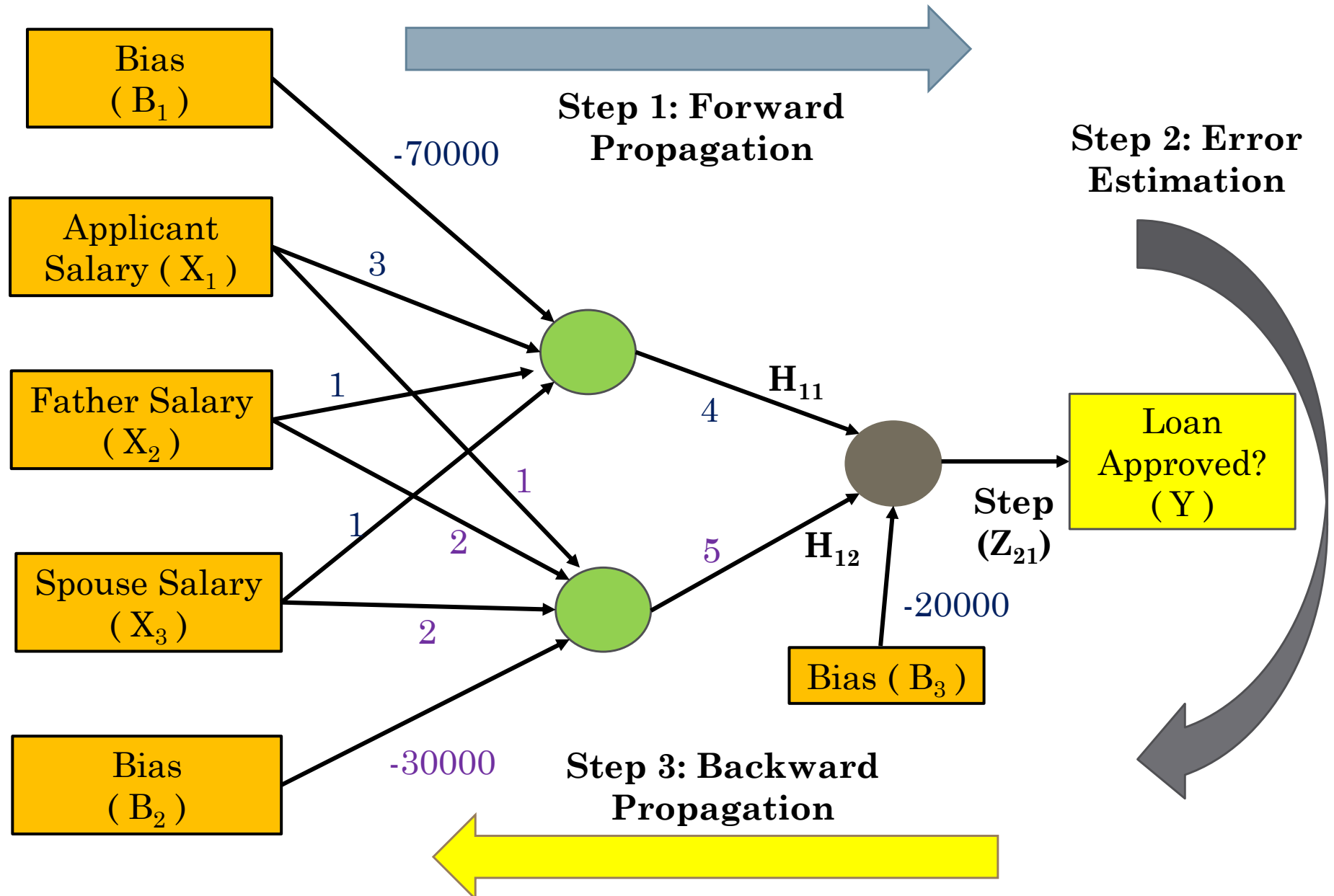


Backward Propagation

$$\text{Error} = \frac{(Y_{\text{true}} - Y_{\text{pred}})^2}{2}$$

Hyperparameters:
Activation
Function
Inputs

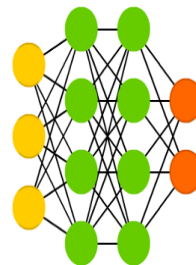
Parameters:
Weights
Biases



Neural Networks

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Deep Feed Forward (DFF)



Perceptron (P)



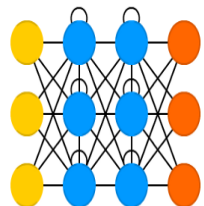
Feed Forward (FF)



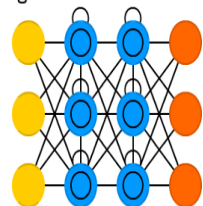
Radial Basis Network (RBF)



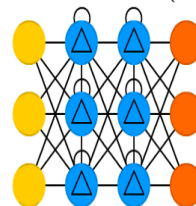
Recurrent Neural Network (RNN)



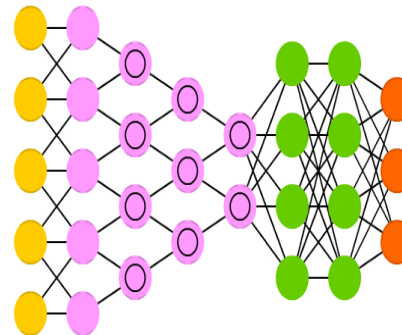
Long / Short Term Memory (LSTM)



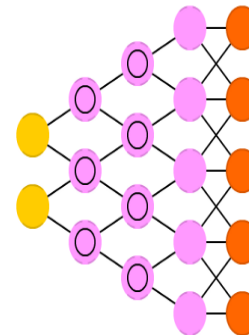
Gated Recurrent Unit (GRU)



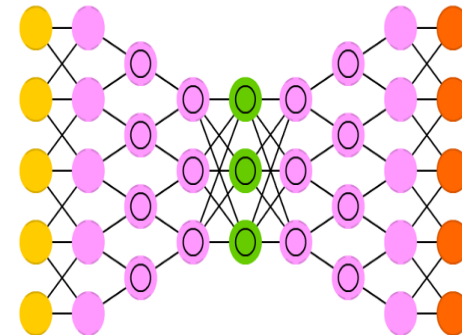
Deep Convolutional Network (DCN)



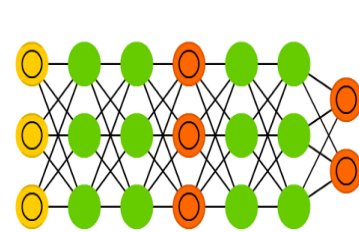
Deconvolutional Network (DN)



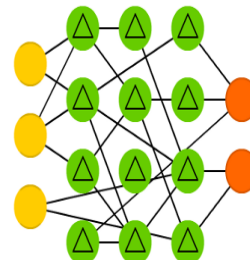
Deep Convolutional Inverse Graphics Network (DCIGN)



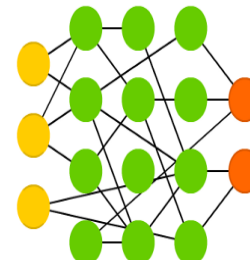
Generative Adversarial Network (GAN)



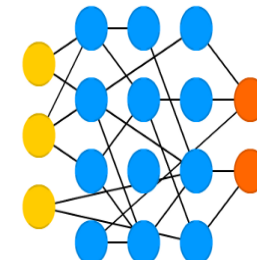
Liquid State Machine (LSM)



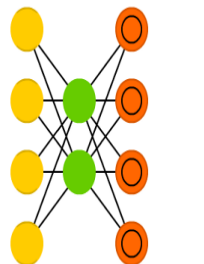
Extreme Learning Machine (ELM)



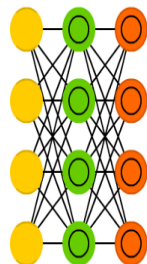
Echo State Network (ESN)



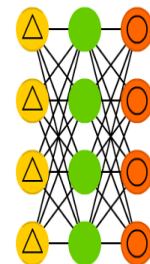
Auto Encoder (AE)



Variational AE (VAE)



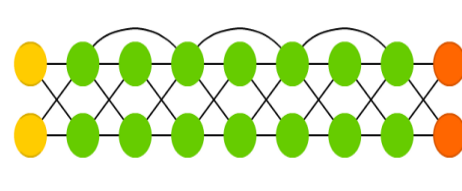
Denoising AE (DAE)



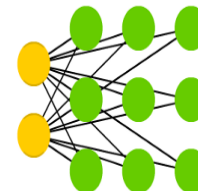
Sparse AE (SAE)



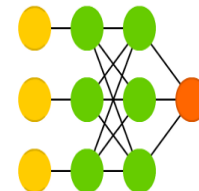
Deep Residual Network (DRN)



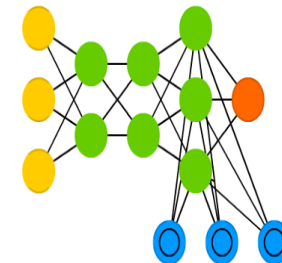
Kohonen Network (KN)



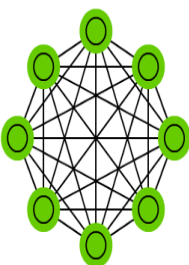
Support Vector Machine (SVM)



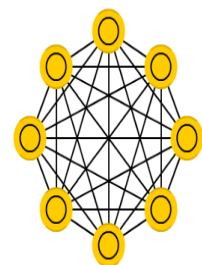
Neural Turing Machine (NTM)



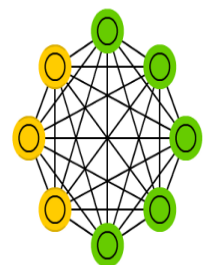
Markov Chain (MC)



Hopfield Network (HN)



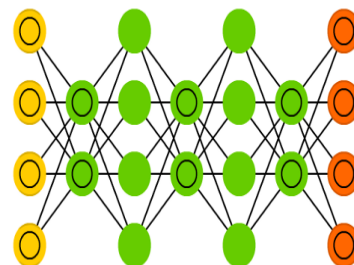
Boltzmann Machine (BM)



Restricted BM (RBM)



Deep Belief Network (DBN)



Quiz

Which of the following things can be updated during backpropagation?

Choose ALL answers that apply.



Weights



Biases

C

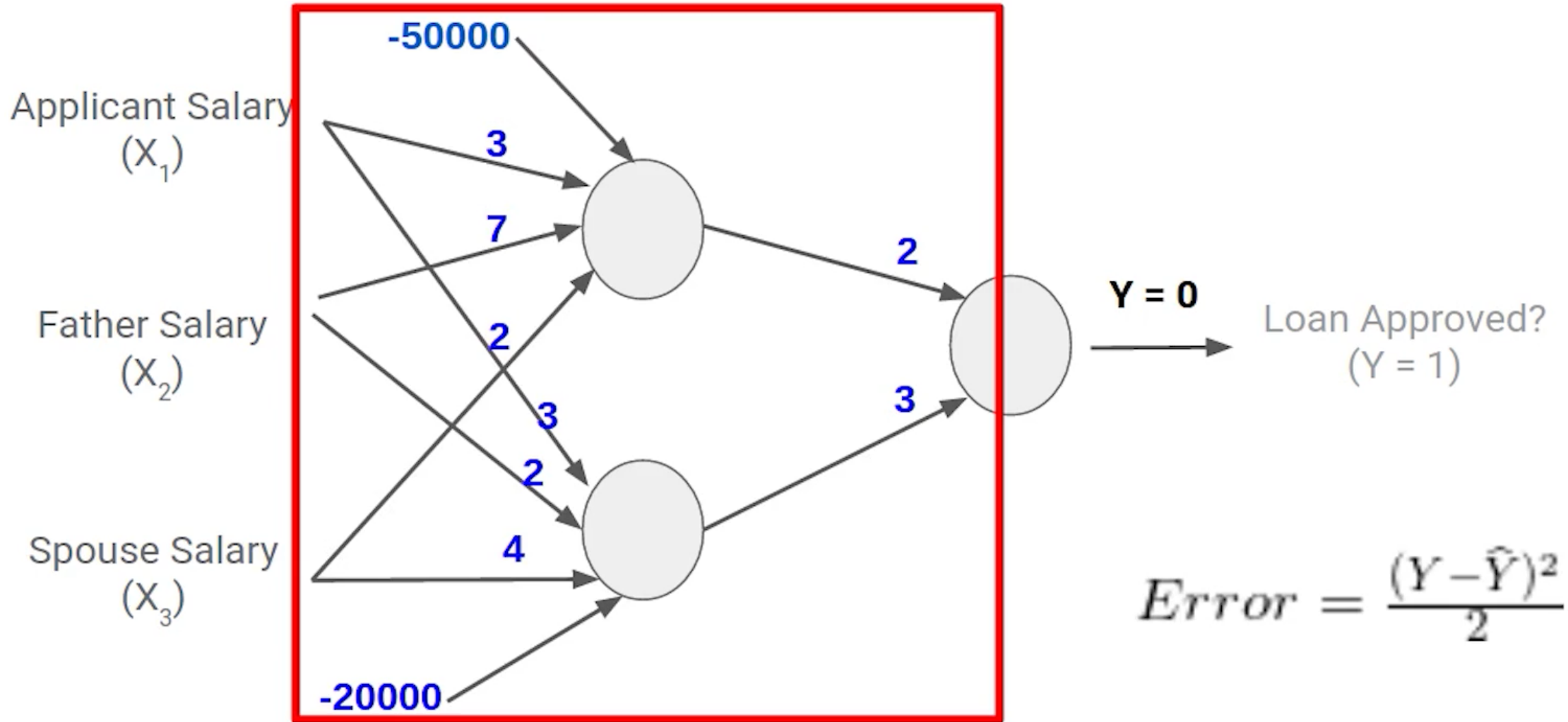
Inputs

D

Actual Target

Gradient Descent Algorithm

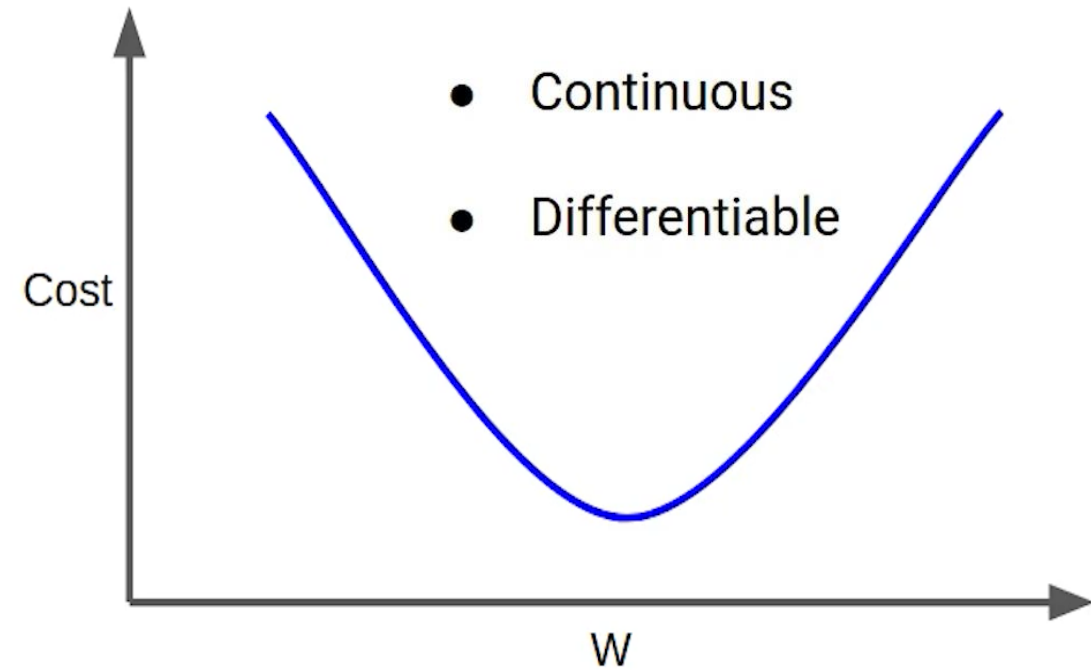
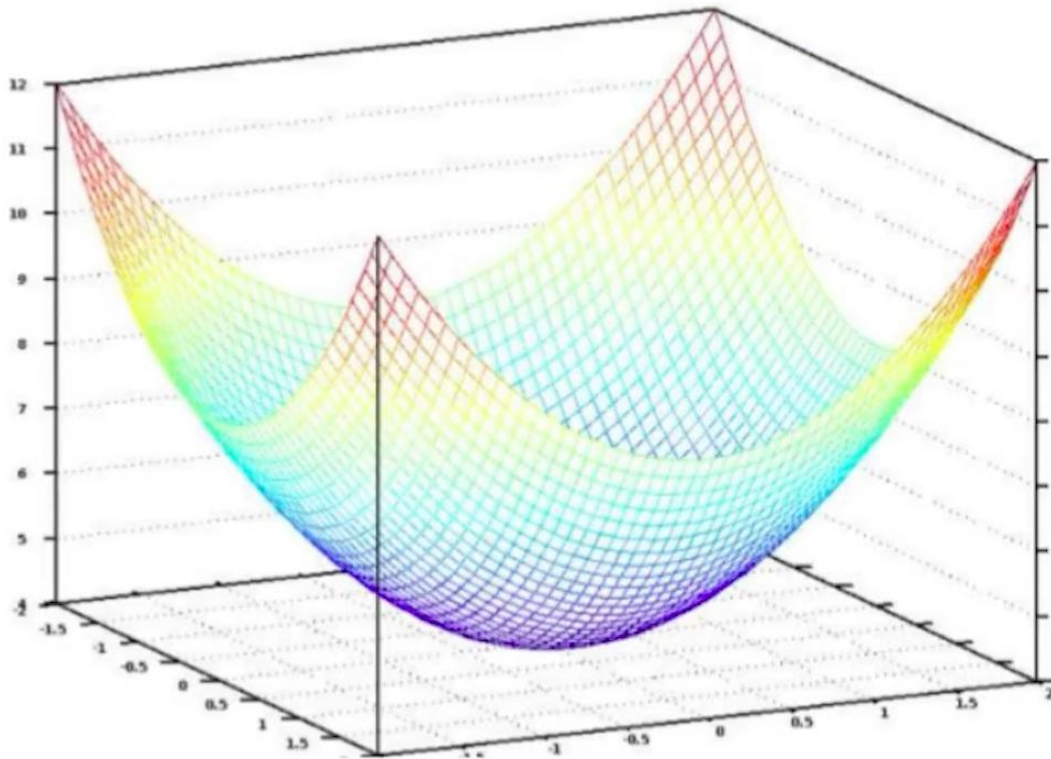
Updating Weights and Bias



What next? Optimize w and b to minimize the error

Updating Weights and Bias

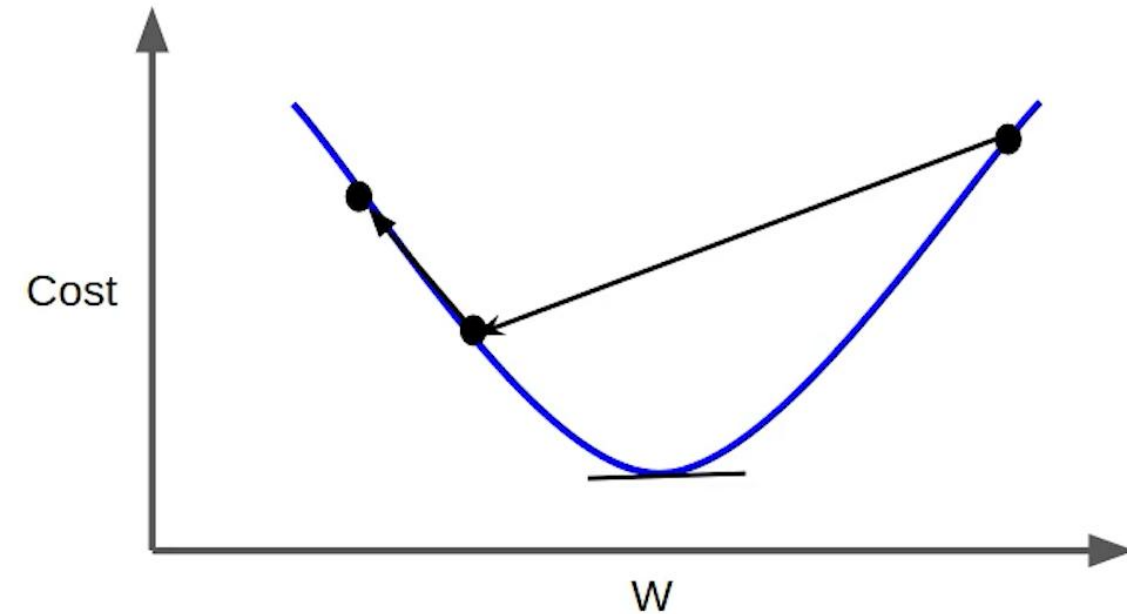
How to update w and b ? Update them randomly



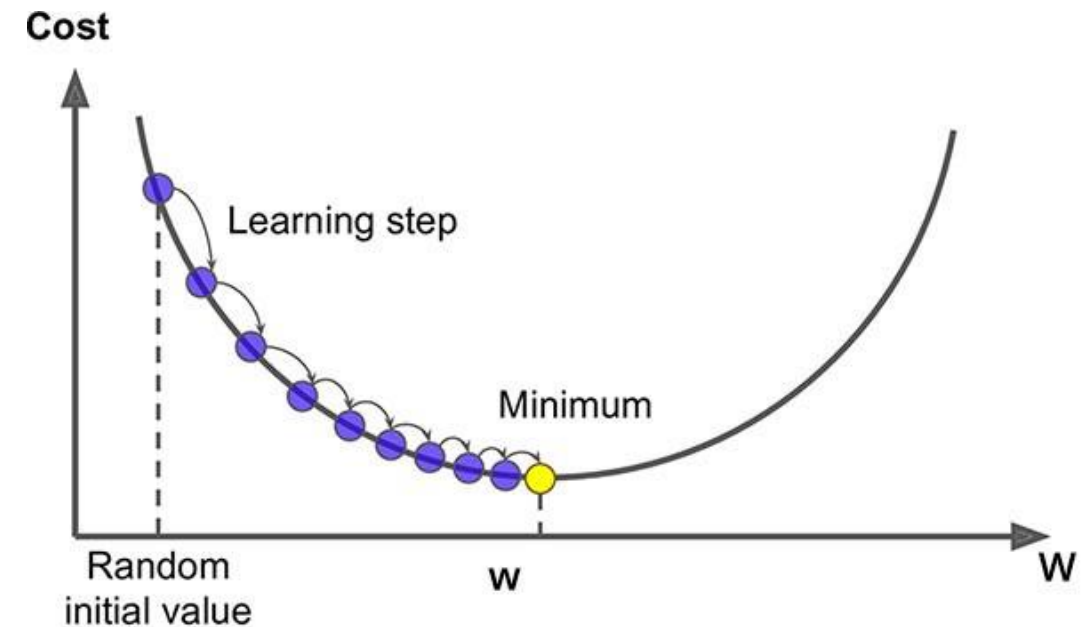
Good Explanation: <https://realpython.com/gradient-descent-algorithm-python/>

Updating Weights and Bias

How to update w and b ? Update them randomly



- In which direction should I move?
- How much should I move?

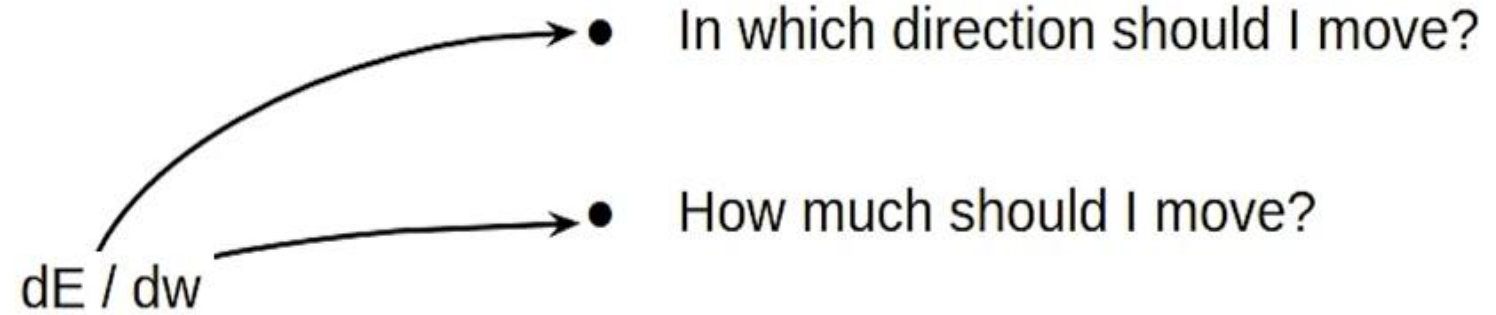


Understanding Gradient Descent

Gradient Descent update equation

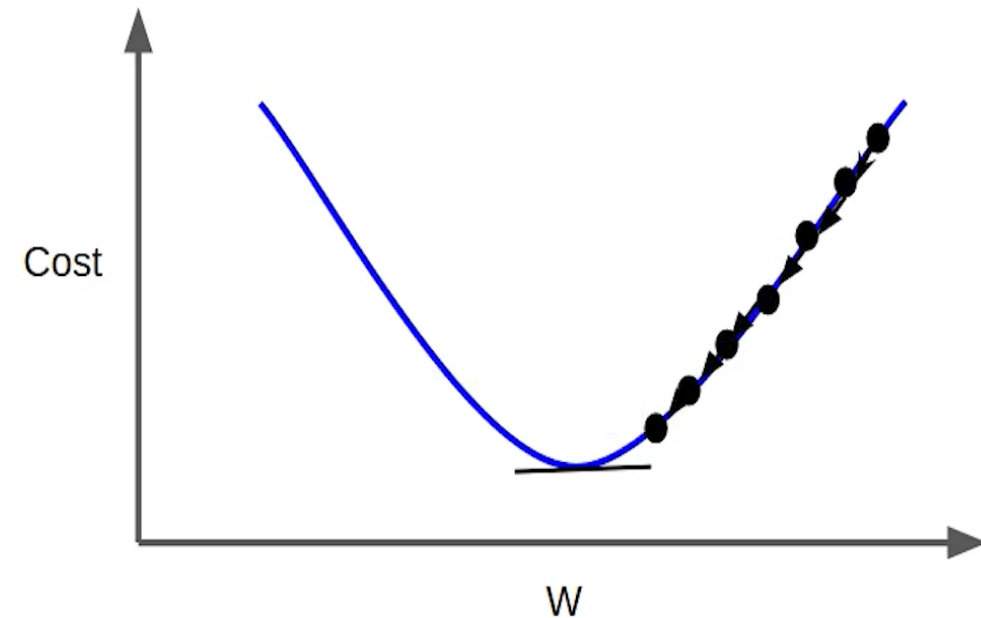
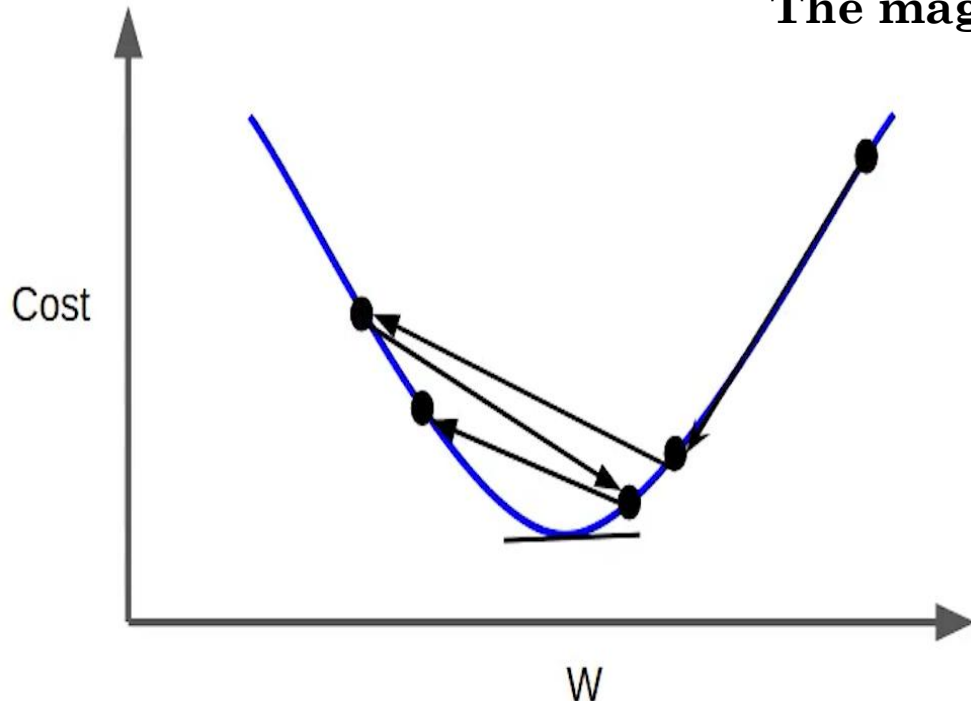
$$w = w - \alpha * dE / dw$$

α is the learning rate



The sign of $\frac{dE}{dw}$ decides the direction in which we should move

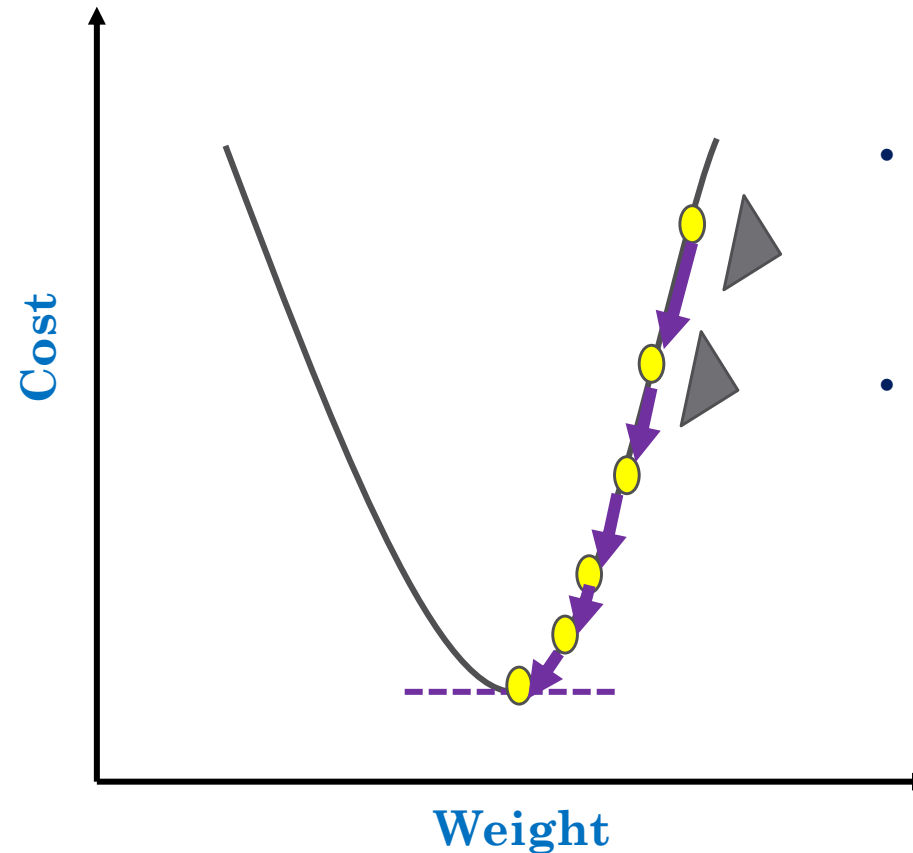
The magnitude of $\frac{dE}{dw}$ decided how much we should move



Steps to Perform Gradient Descent

Steps:

1. Take current value of weight (w) and bias (b) and calculate Cost.
2. Take step in the steepest downhill direction.
3. Repeat step 1 and 2 until minima is achieved.



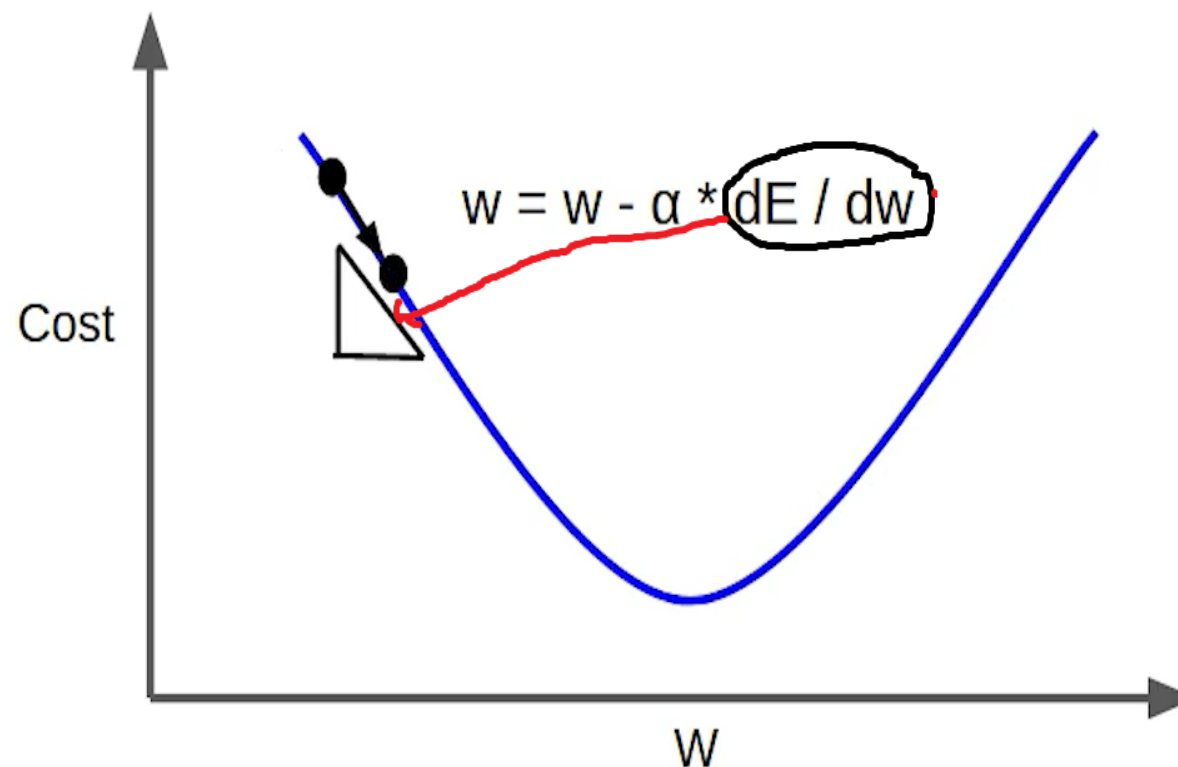
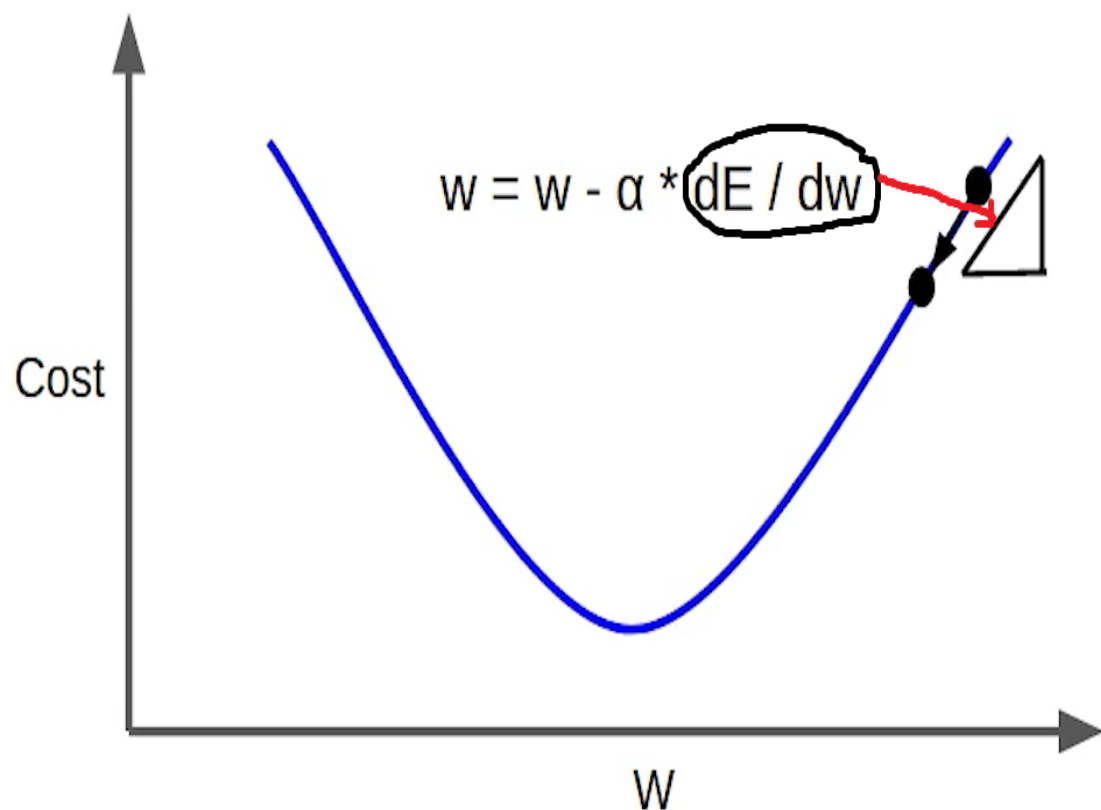
- Calculate the slope at the first point.
- Use Gradient Descent Update Equation:

$$w = w - \alpha * \frac{dE}{dw}$$

- Repeat same process until minimal value of cost function reaches.

Steps to Perform Gradient Descent

- Take current value of w and b
 - Take a step in the steepest downhill direction
 - Repeat the previous step until minima is achieved
1. Error is not updating
 2. Number of iterations is reached



What is Gradient Descent?



Gradient Descent is an optimization algorithm used to minimize a cost function.



It iteratively adjusts parameters in the opposite direction of the gradient to find the minimum point.

Function and Its Derivative

Let's minimize:

$$f(x) = (x - 3)^2$$

Derivative:

$$f'(x) = 2(x - 3)$$

Minimum is at $x = 3$

Gradient Descent Update Rule



Update formula:



$$x_{\text{new}} = x_{\text{old}} - \eta * f'(x)$$



η is the learning rate (step size)



We repeat this until the slope is nearly zero

Step-by-Step Example

Starting at $x = 0$, with
learning rate $\eta = 0.1$:

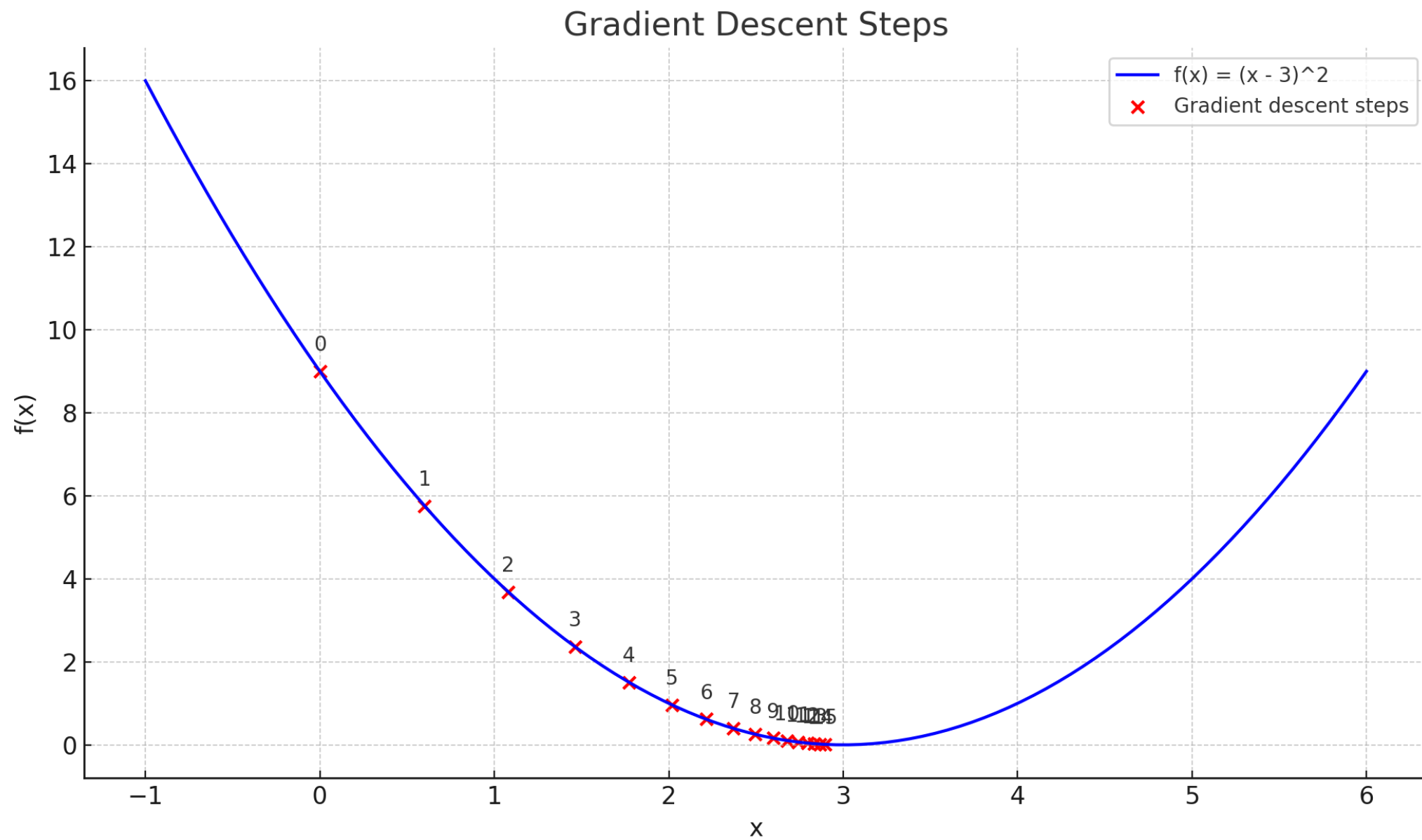
Iteration 1: $x = 0.6$

Iteration 2: $x = 1.08$

Iteration 3: $x = 1.464$

... until $x \approx 3$

Gradient Descent Visualization



Quiz

What are the stopping criteria for gradient descent algorithm?

Choose only ONE best answer.

A

Error is not updating anymore

B

Number of iterations is reached



Both of the above

D

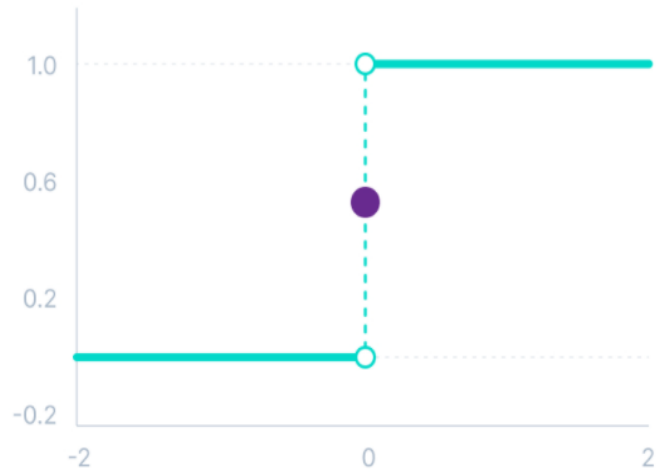
None of the above

BREAK

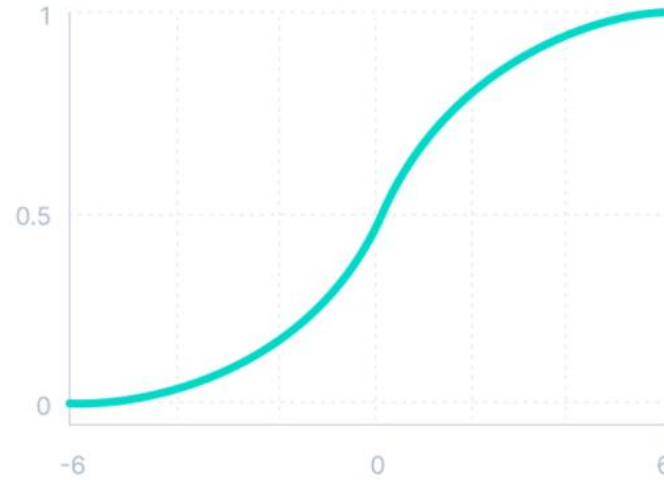
Activation Functions

Activation Functions

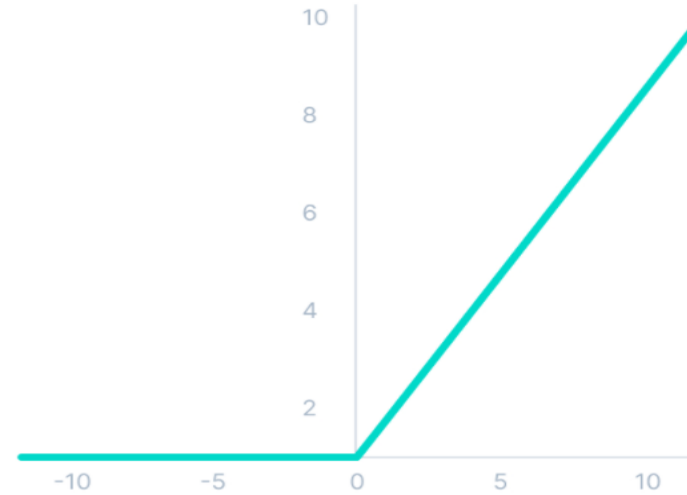
Binary Step Function



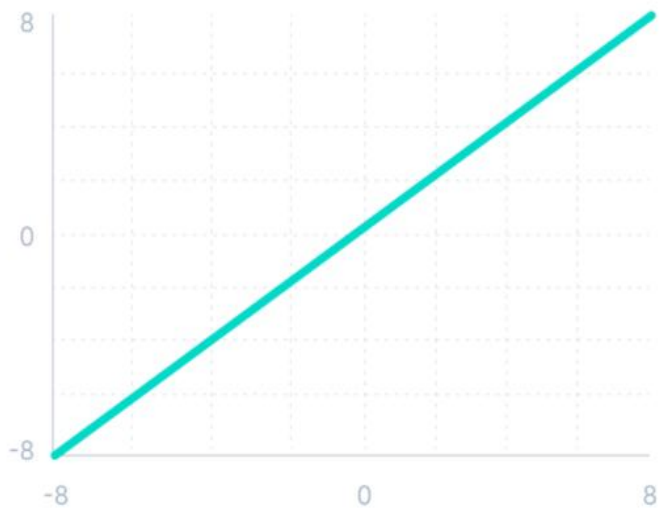
Sigmoid / Logistic



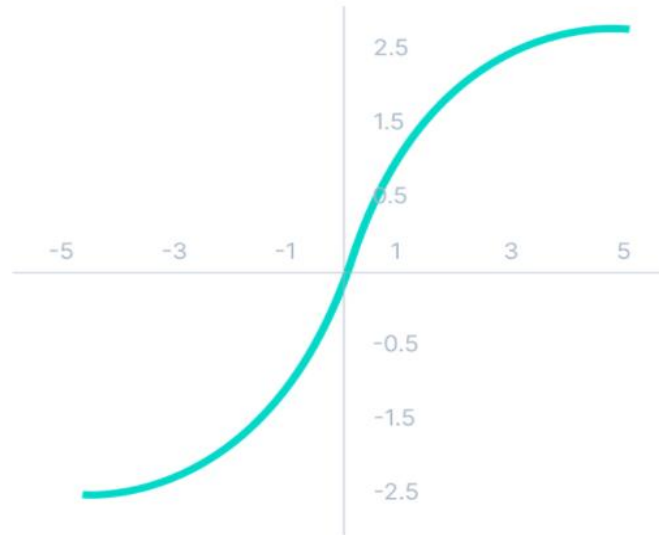
ReLU



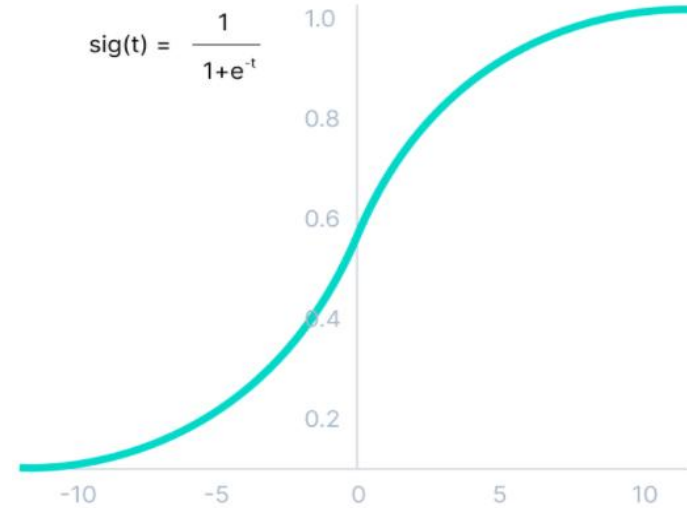
Linear Activation Function



Tanh



Softmax

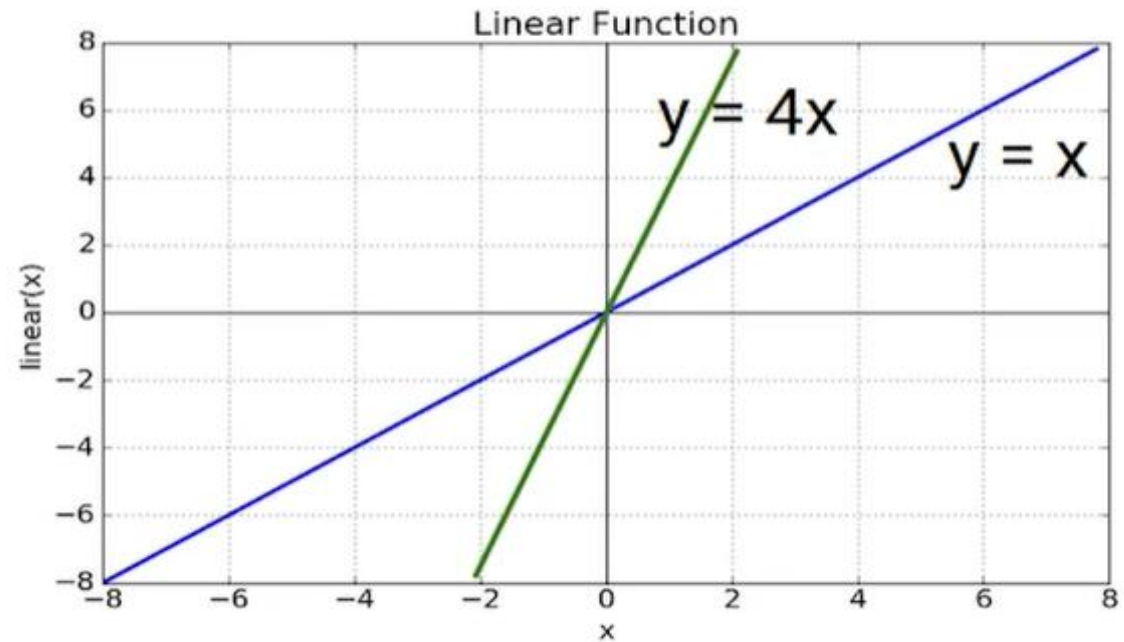


$$\text{sig}(t) = \frac{1}{1+e^{-t}}$$

Linear Activation Function

- Linear Activation Function:
- Graph of Linear Function:
- Input Range: $(-\infty \text{ to } \infty)$
- Output Range: $(-\infty \text{ to } \infty)$

$$y = ax$$

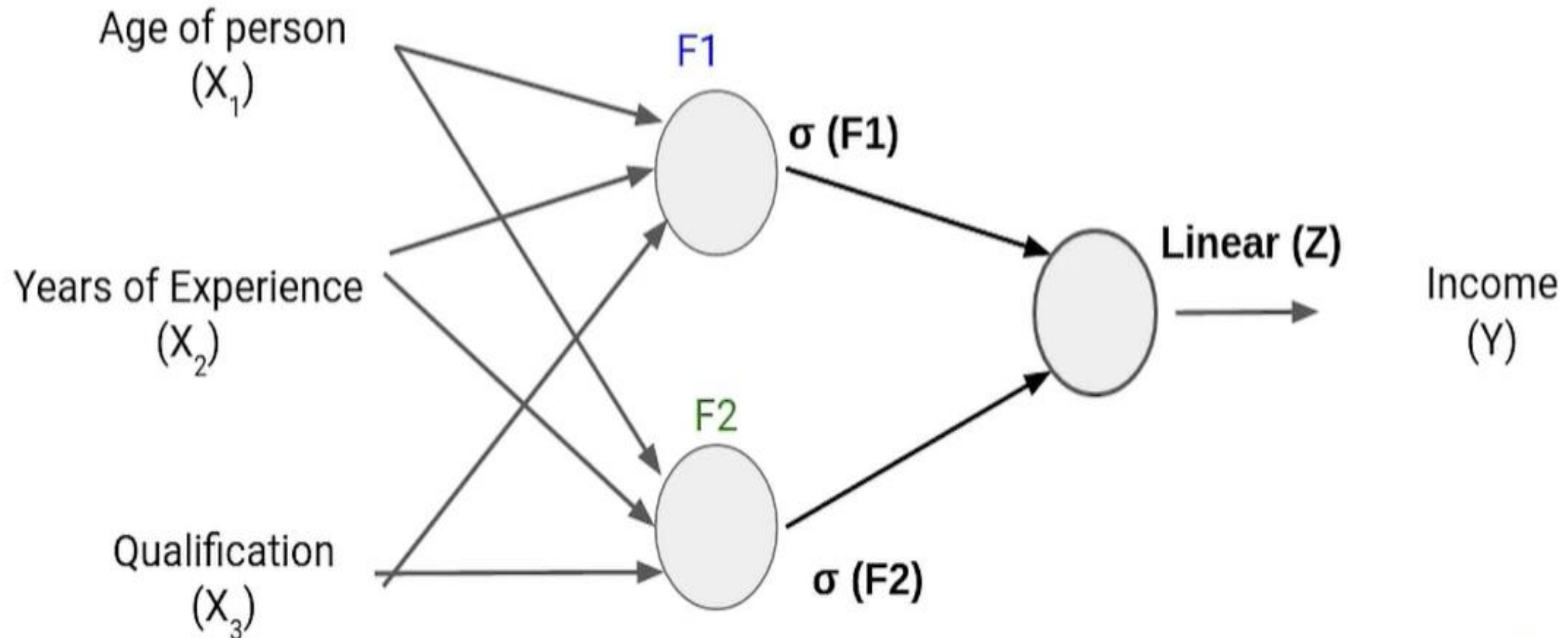


Linear Activation Function

- Linear Activation Function:
- Linear Activation Function derivative:

$$y = ax$$

$$\frac{dy}{dx} = a$$



Quiz

What is the input range of a linear activation function?

Choose only ONE best answer.

A 0 to 1

B 0 to infinity

C -infinity to 0



-infinity to infinity

What is the output range of a linear activation function?

Choose only ONE best answer.

A 0 to 1

B 0 to infinity

C -infinity to 0

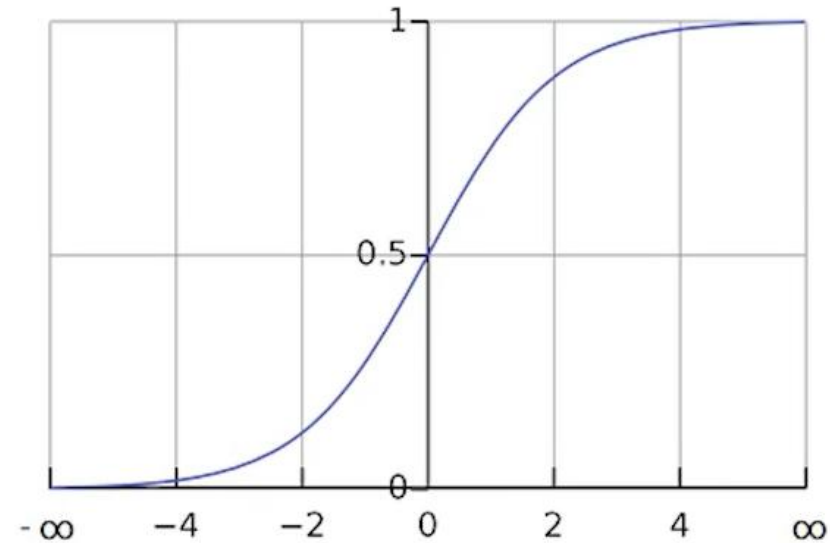


-infinity to infinity

Sigmoid Activation Function

- Sigmoid Function Formula:
$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

- Sigmoid Function Graph:



- Input Range: $(-\infty \text{ to } \infty)$
- Output Range: $(0 \text{ to } 1)$
- Continuous and Differentiable

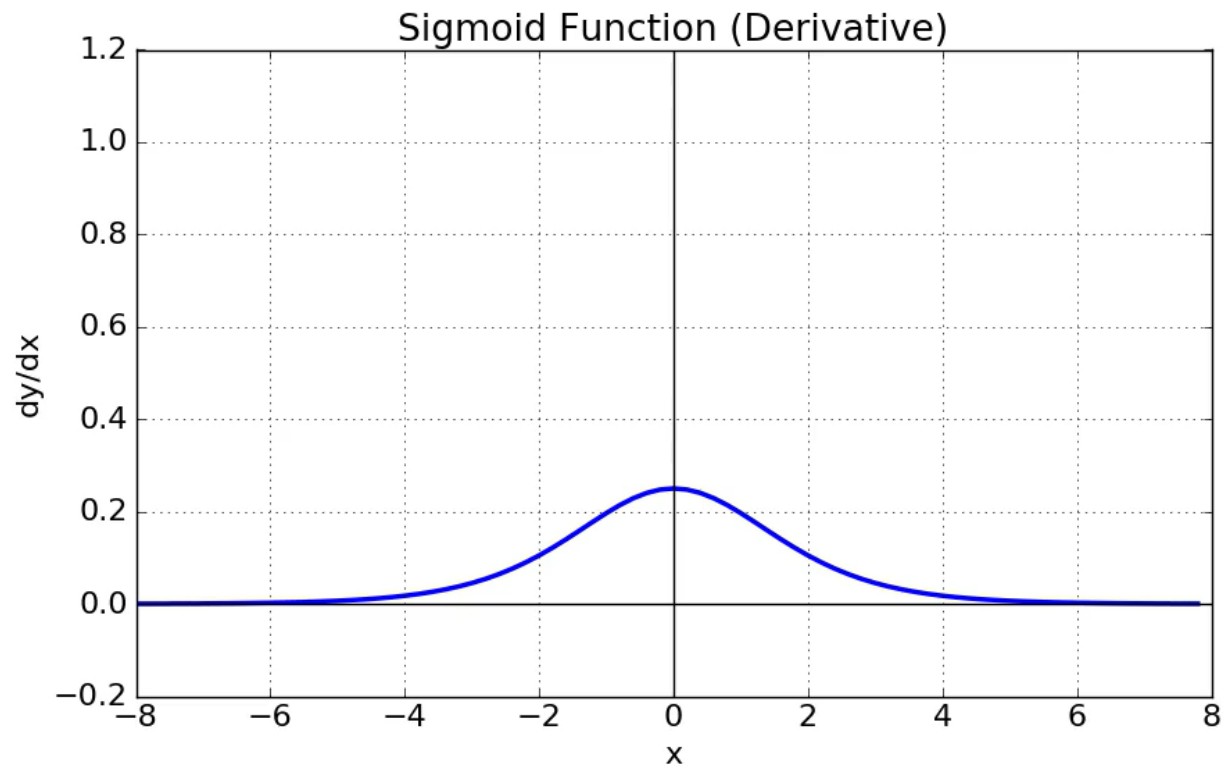
Sigmoid Activation Function

- Sigmoid Function:

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

- Sigmoid Function derivative:

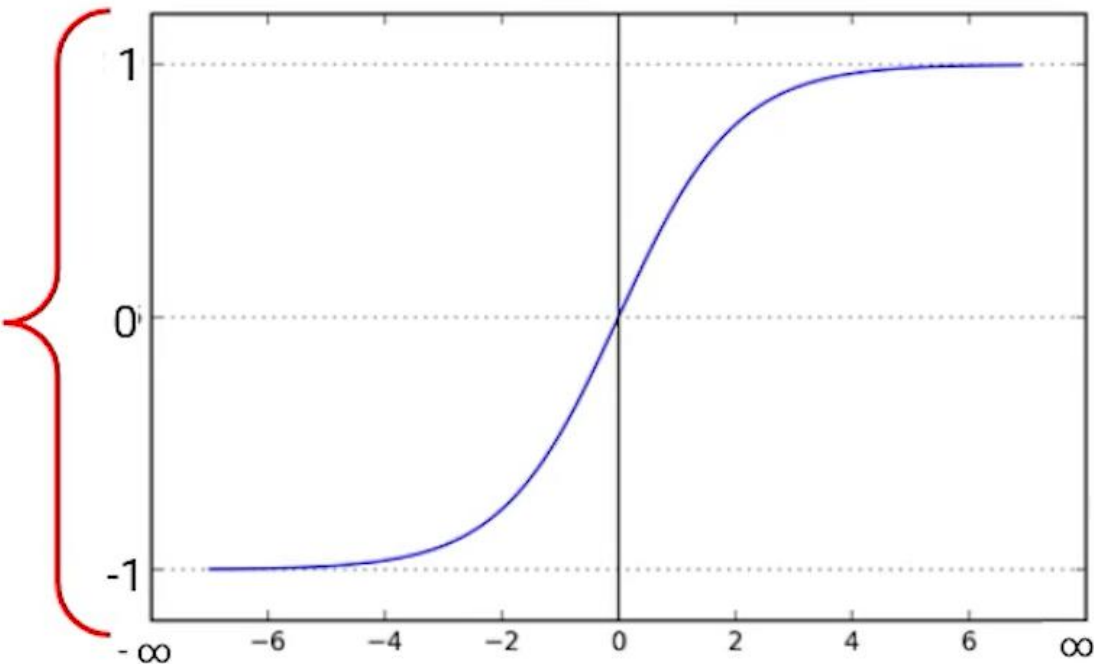
$$\frac{d\sigma}{dx} = \sigma(x) (1 - \sigma(x))$$



Tanh or Scaled Sigmoid Activation Function

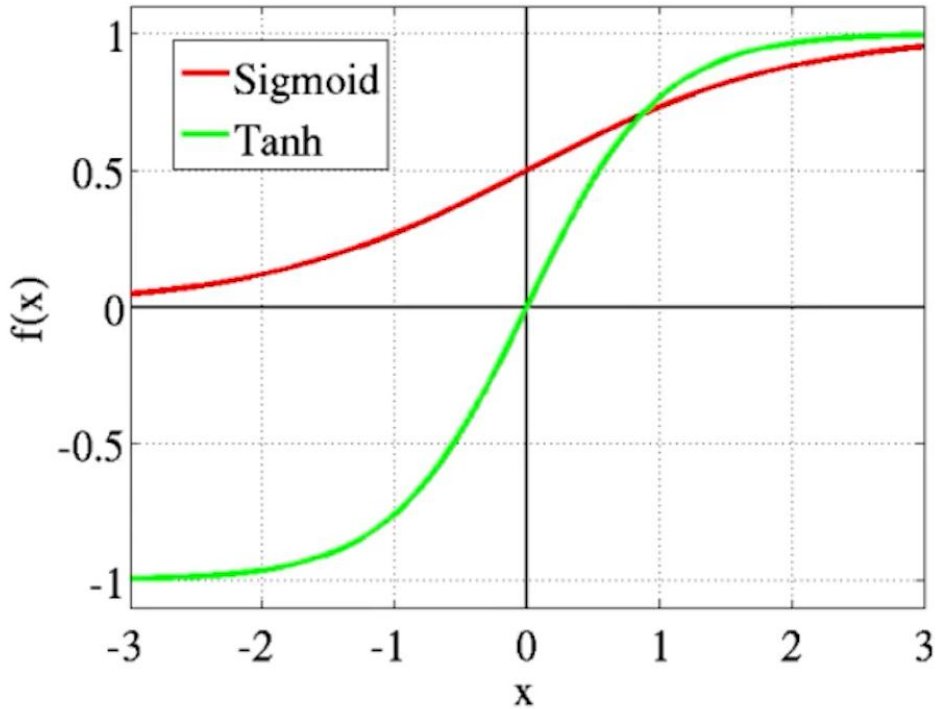
- Tanh Activation Function: $\tanh(x) = \frac{2}{1+e^{-2x}} - 1$

$$\sigma(x) = \frac{1}{1+e^{-(x)}} \quad \tanh(x) = 2 \sigma(2x) - 1$$



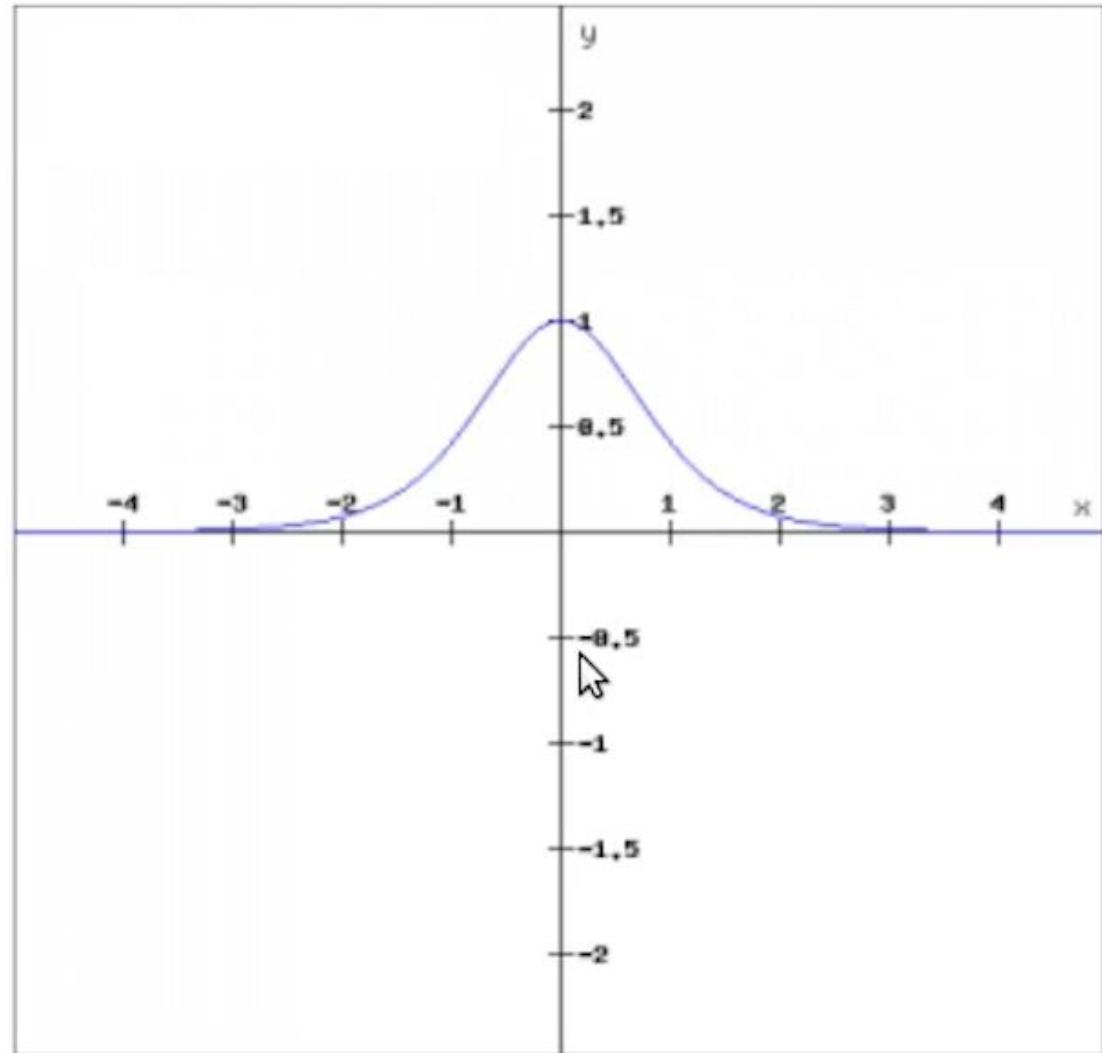
- Input Range: $(-\infty \text{ to } \infty)$
- Output Range: $(-1 \text{ to } 1)$

Tanh or Scaled Sigmoid Activation Function



- Tanh Function derivative:

$$\frac{d \tanh}{d x} = 1 - \tanh^2(x)$$



Illustration

<http://playground.tensorflow.org/#activation=tanh&batchSize=10&dataset=circle®Dataset=reg-plane&learningRate=0.03®ularizationRate=0&noise=0&networkShape=4,2&seed=0.68420&showTestData=false&discretize=false&percTrainData=50&x=true&y=true&xTimesY=false&xSquared=false&ySquared=false&cosX=false&sinX=false&cosY=false&sinY=false&collectStats=false&problem=classification&initZero=false&hideText=false>

Quiz

What is the output range of a sigmoid activation function?

Choose only ONE best answer.



0 to 1

B

0 to infinity

C

-infinity to 0

D

-infinity to infinity

What is the output range of tanh activation function?

Choose only ONE best answer.

A

0 to 1

B

0 to infinity



-1 to 1

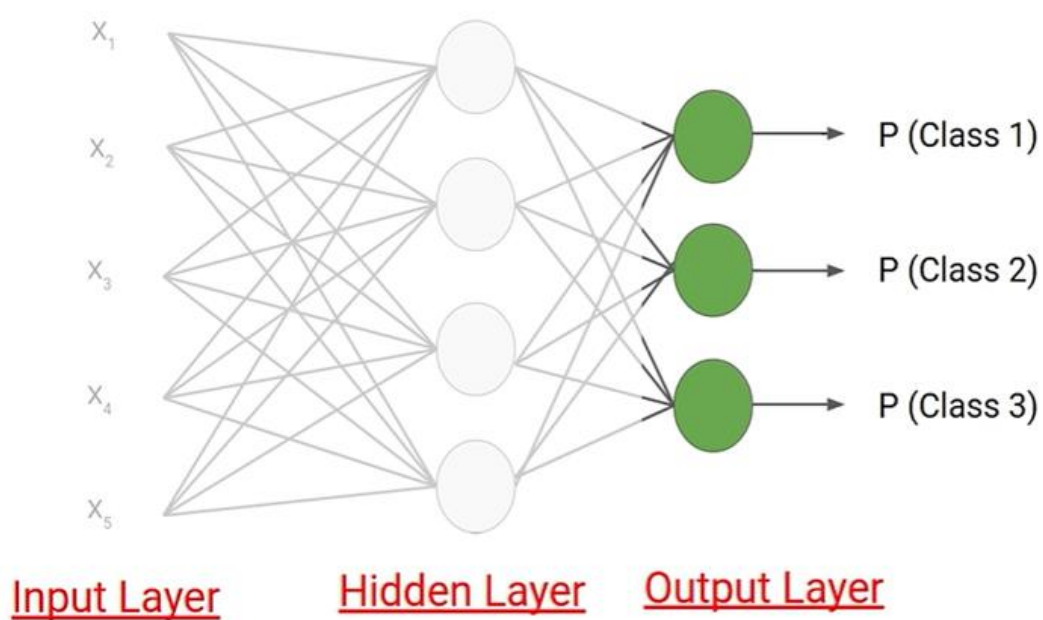
D

-infinity to infinity

Softmax Activation Function

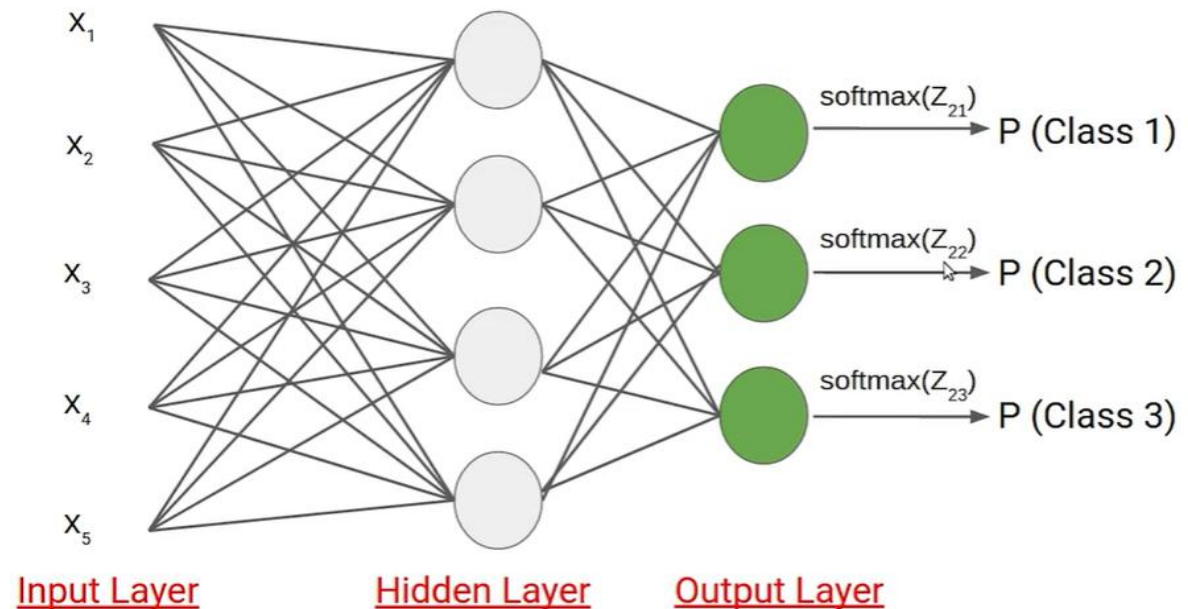
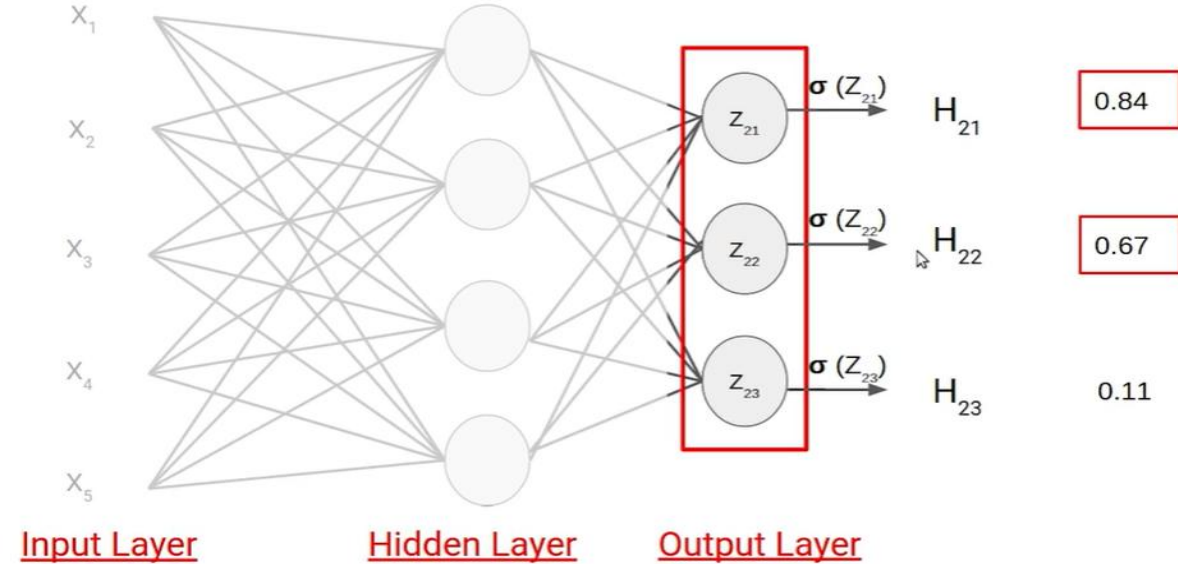
Feature X1	Feature X2	Feature X3	Feature X4	Feature X5	Target
1	22	569	35	0	Class 1
1	7	351	75	1	Class 2
1	45	451	542	1	Class 2
1	5	572	8	0	Class 1
0	22	565	44	1	Class 3
0	24	243	546	1	Class 3
1	78	953	42	0	Class 2

Softmax Activation Function

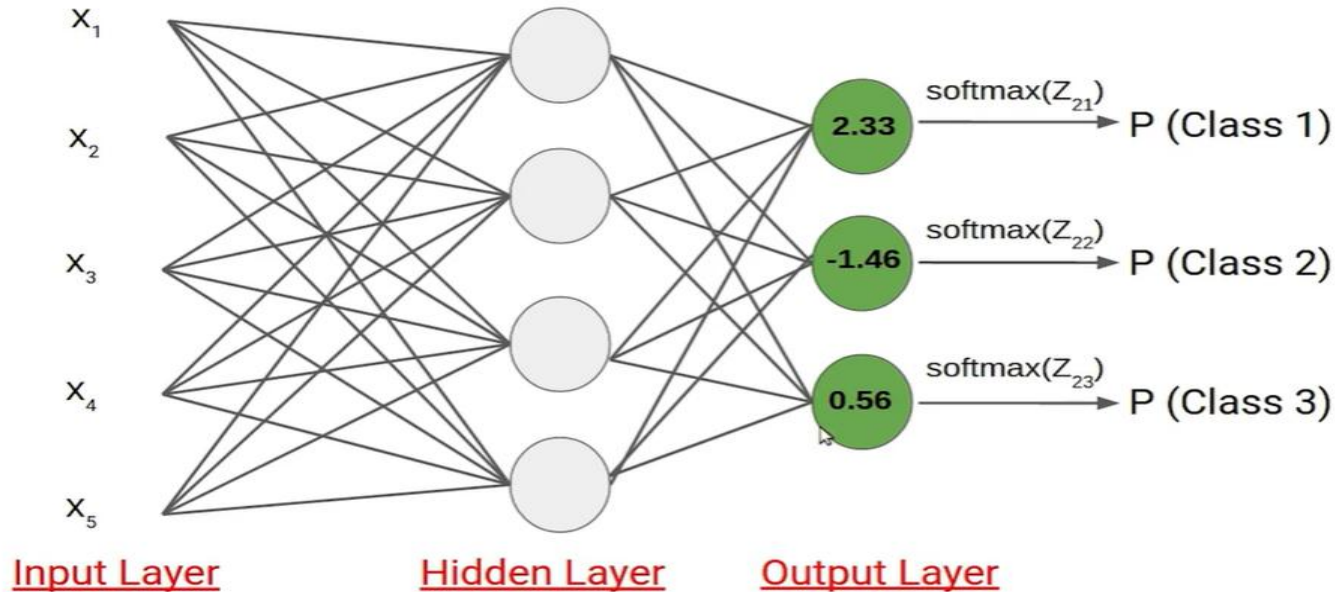


- Returns probability for each class
- Sigmoid Activation Function:
- Non-linear activation function
- Probability of all classes

$$\text{softmax}(z_i) = \frac{\exp(z_i)}{\sum_j \exp(z_j)}$$



Softmax Activation Function



$$\text{softmax}(z_i) = \frac{\exp(z_i)}{\sum_j \exp(z_j)}$$

Example :

$$2.33 \rightarrow P(\text{Class 1}) = \frac{\exp(2.33)}{\exp(2.33) + \exp(-1.46) + \exp(0.56)} = 0.83827314$$

$$-1.46 \rightarrow P(\text{Class 2}) = \frac{\exp(-1.46)}{\exp(2.33) + \exp(-1.46) + \exp(0.56)} = 0.01894129$$

$$0.56 \rightarrow P(\text{Class 3}) = \frac{\exp(0.56)}{\exp(2.33) + \exp(-1.46) + \exp(0.56)} = 0.14278557$$

Quiz

The softmax activation function is used in the case of four classification problems.

Choose only ONE best answer.



True

B

False

Sum of probabilities of the output in case of softmax activation is 1.

Choose only ONE best answer.



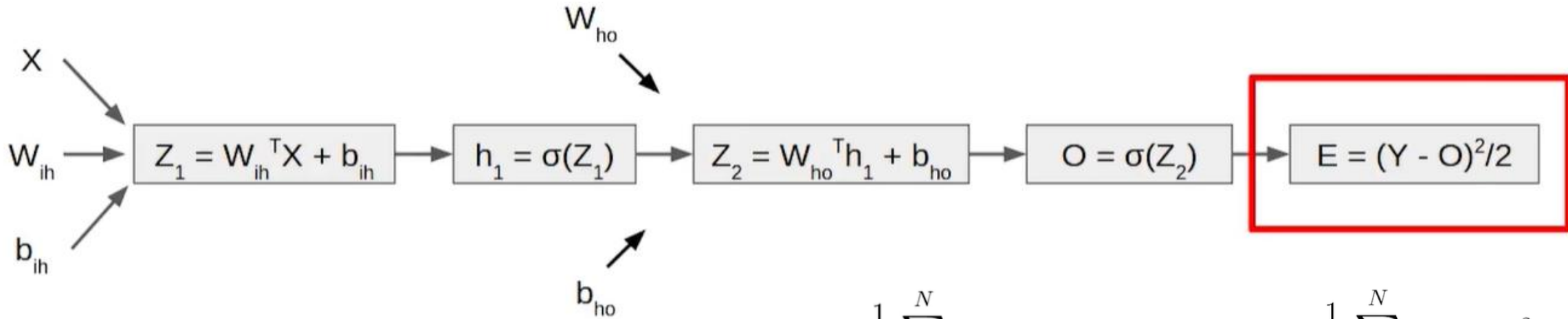
True

B

False

Loss Functions

Loss Function for Regression



- Mean Absolute Error
- Mean Squared Error
- Root Mean Squared Error

$$RMSE = \sqrt{\frac{\sum_{i=1}^N (Predicted_i - Actual_i)^2}{N}}$$

$$MSE = 44$$

$$RMSE = 6.63$$

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i|$$

Actual Values	Predicted Values	Absolute Error
19	28	9
37	33	4
25	20	5
9	16	7
22	15	7

$$MAE = 6.4$$

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

Actual Values	Predicted Values	Squared Error
19	28	81
37	33	16
25	20	25
9	16	49
22	15	49

$$MSE = 44$$

Quiz

Which of the following is / are loss function(s) for a regression problem?

Choose only ONE best answer.

A Mean Squared Error

B Mean Absolute Error

C Root Mean Squared Error



All of the above

Loss Function for Classification

Binary Cross Entropy/Log Loss: It is the negative average of the log of corrected predicted probabilities

ID	Actual	Predicted probabilities	Corrected Probabilities	Log
ID6	1	0.94	0.94	-0.0268721464
ID1	1	0.90	0.90	-0.0457574906
ID7	1	0.78	0.78	-0.1079053973
ID8	0	0.56	0.44	-0.3565473235
ID2	0	0.51	0.49	-0.30980392
ID3	1	0.47	0.47	-0.3279021421
ID4	1	0.32	0.32	-0.4948500217
ID5	0	0.10	0.90	-0.0457574906

$$- \frac{1}{N} \sum_{i=1}^N (\log(p_i)) \quad \text{Log Loss} = 0.2144$$

- p_i is probability of 1
- $1-p_i$ is probability of 0

$$\text{Log loss} = \frac{1}{N} \sum_{i=1}^N - (y_i * \log(p_i) + (1-y_i) * \log(1-p_i))$$

$$\text{Log loss} = - \frac{1}{N} \sum_{i=1}^N (y_{i1} * \log(p_{i1}) + y_{i2} * \log(p_{i2}))$$

- When the actual class is 1:
 - $y_{i1} = 1, y_{i2} = 0$
- When the actual class is 0:
 - $y_{i1} = 0, y_{i2} = 1$
- p_{i1} and p_{i2} : probability of first and second class

Categorical Cross Entropy:

$$\text{logloss} = - \frac{1}{N} \sum_i^N \sum_j^M y_{ij} \log(p_{ij})$$

- N is the number of rows
- M is the number of classes

Quiz

A multiclass classification problem is one where we have two categories in the target variable.

Choose only ONE best answer.

A

True



False

Why do we take the log of the corrected probabilities?

Choose only ONE best answer.

A

It offers less penalty for small differences

B

It offers more penalty for large differences



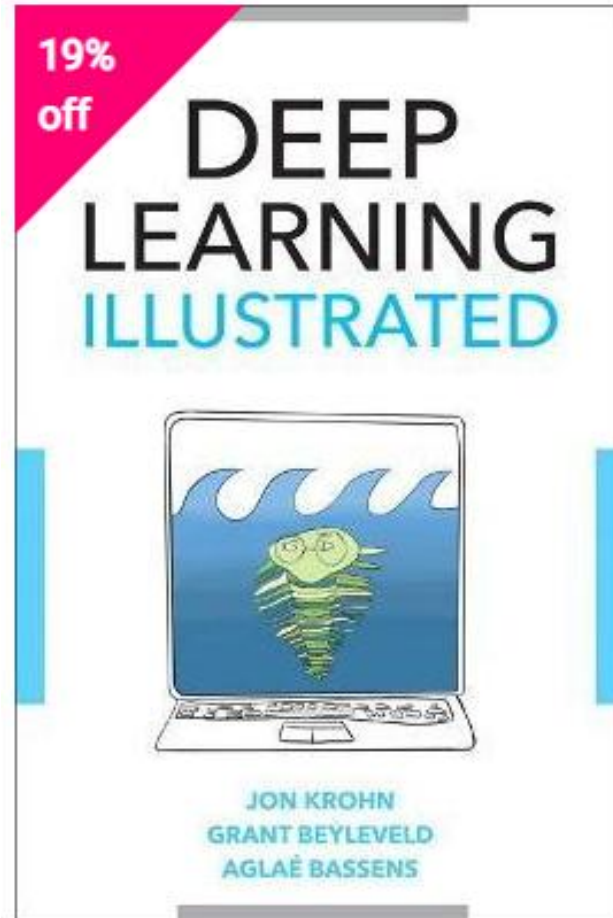
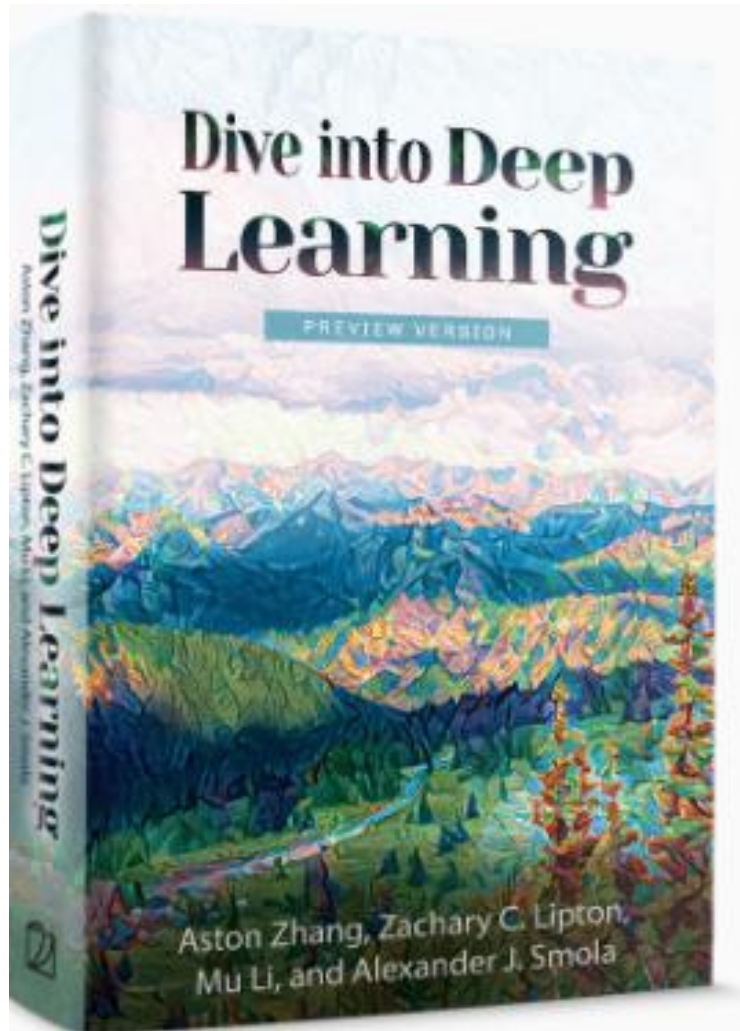
Both of the these

D

None of these

Good Books

<https://d2l.ai/index.html>



Thank You
for
Attention

